

Chapter 2

Stock Ownership and Inflation Expectations: Evidence from the 2021–2025 Inflation Surge

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Abstract

The gap between stockholders' and non-stockholders' inflation expectations widened during the prolonged 2021–2025 inflation episode and grew further as inflation receded. Using the University of Michigan Survey of Consumers in a difference-in-differences design, I estimate that stockholders' one-year-ahead expectations rose 1.78 percentage points less than non-stockholders', alongside a relative improvement in forecast accuracy. The widening is concentrated in the upper tail: the estimated difference is 4.67 points at the ninetieth percentile and near zero at the twenty-fifth. No comparable widening is detected around the shorter 2008 inflation spike. I develop a two-signal model of behavioural inattention in which households form forecasts from everyday prices and aggregate financial information. Stock ownership lowers the cost of attending to financial information, reducing the influence of accumulated price experience on forecasts. A calibration disciplined by group means and forecast dispersions shows how fixed differences in attention costs translate a changing inflation environment into a widening expectations gap.

Keywords: inflation expectations, stock ownership, behavioural inattention, perceptual bias, difference-in-differences, household heterogeneity.

JEL: E31; D84; G51; D83.

2.1 Introduction

Between 2021 and 2025 the United States lived through its most sustained inflation in four decades (Bernanke and Blanchard, 2023; Ball et al., 2022). Households raised their expectations, but not together. What separated those who stayed near the aggregate figures from those who ran far above them was not income or schooling alone. It was whether they owned stocks. In the University of Michigan Survey of Consumers, stockholders and non-stockholders had differed by 0.76 percentage points for two decades. Across the surge that gap more than tripled, and extreme beliefs became markedly more common among non-stockholders: 35 percent of them expected inflation above ten percent, against 22 percent of stockholders.

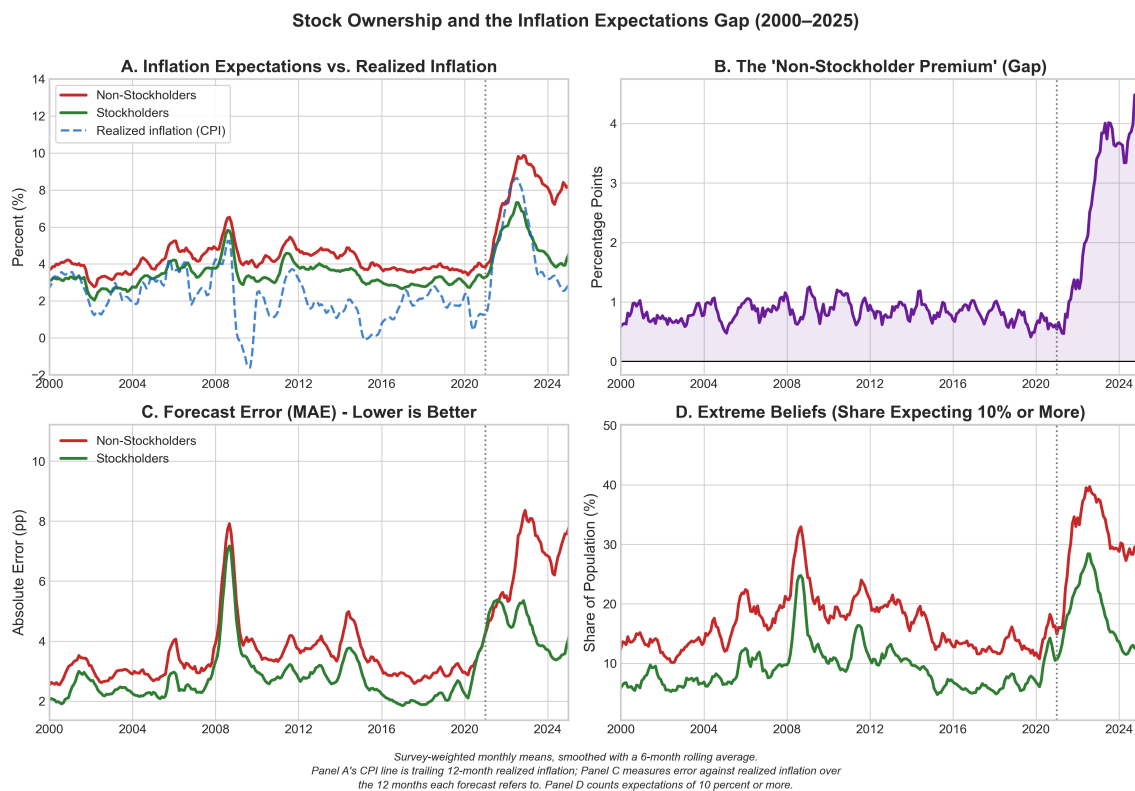


Figure 1: Stock ownership and the inflation-expectations gap, 2000–2025. Panel A: mean one-year-ahead expectations of stockholders and non-stockholders against trailing twelve-month realized CPI inflation. Panel B: the stockholder–non-stockholder expectations gap. Panel C: mean absolute forecast errors, measured against realized inflation over the twelve months each forecast refers to. Panel D: share of each group expecting inflation of ten percent or more. Series are survey-weighted monthly means smoothed with a six-month rolling average. Data: Michigan Survey of Consumers; the vertical line marks January 2021.

Selection is the obvious answer. Stockholders are richer and better educated.¹ Those households may forecast better for reasons that have nothing to do with the stock market. Fixed traits can account for a standing difference in levels. To account for a change, they would have to interact with the shock itself, and the 2008 episode gives no sign of that: inflation volatility was comparable and the gap did not open. Timing is the puzzle this paper takes up.

I use the Michigan survey, a monthly repeated cross-section that records both one-year-ahead inflation expectations and stock ownership, and compare the two groups before and after January 2021. Year fixed effects absorb whatever moved both groups together, so the estimate rests on their differential movement. Survey composition shifts over time, so I let each demographic group take its own post-2021 shift; the headline is then net of any tendency for women, younger, less-educated or lower-income households to respond differently to the shock. Stockholders' expectations rose 1.78 percentage points less than comparable non-stockholders'. Adding controls moves the estimate from 2.48 in its rawest form to 2.04 with socioeconomic controls and 1.78 in the most demanding specification. In terms of accuracy, stockholders' absolute forecast errors fell 1.38 points more than non-stockholders', and their upward bias 1.50 points more.

The contribution is to show that a belief cleavage long read as a fixed feature of the cross-section is state-dependent. Work on experienced prices explains why households disagree at a point in time ([Malmendier and Nagel, 2016](#); [D'Acunto et al., 2021a](#)); work on inattention explains why some follow aggregate news and others do not ([Sims, 2003](#); [Mankiw and Reis, 2002](#); [Gabaix, 2014](#)). Neither asks what happens to the distance between two groups when an episode drags on. [Ahn and Xie \(2024\)](#) establish that stockholders forecast inflation more accurately and revise more responsively. I take that observable margin and ask how the structure of disagreement responds as inflation persists, which links a household-finance question, who holds stocks, to a macroeconomic one, how expectations are formed.

Three patterns are consistent with an information-access interpretation. The stockholder effect is larger among women and lower-income households, and gender moderation persists when education and income moderation enter jointly. This pattern is consistent with differences

¹In this sample 55 percent of stockholders are college graduates against 22 percent of non-stockholders, and 54 percent sit in the top two quintiles of the survey's income distribution against 14 percent.

in exposure to salient prices, but does not isolate exposure from financial knowledge or attention costs. The consumer-price accuracy advantage is not accompanied by a detectable improvement in home-price forecast accuracy. The effect is concentrated in the upper tail: it is near zero at the 25th percentile, -0.71 at the median, -2.50 at the 75th and -4.67 at the 90th. The median differs from zero under survey-month clustering but not under the year-cluster sensitivity; the upper-tail estimates remain negative under both. Winsorising high responses substantially reduces the average effect. The widening therefore concerns high forecasts especially, rather than a uniform shift across the distribution.

The same gradient runs within stockholders, where those holding larger positions revised both expectations and errors down more steeply as the surge wore on, and a pre-2021 placebo produces no such gradient. Set against 2008, the later episode ran five times longer, twenty-six consecutive months above four percent against five, while the salient food-and-energy prices the mechanism runs through averaged much the same in both spells, 14.02 percent against 14.97. The gap then kept widening through 2023 and 2024 as inflation itself receded.

A two-signal model of behavioural inattention organises this. Households read inflation from two sources: the prices they meet in daily life, free to observe but a poor guide to the headline rate, and macro-financial news, the better guide but costly to follow. Owning stocks lowers the cost of the second, since monitoring a portfolio brings macroeconomic information within reach anyway. As experienced prices become a worse guide, households already paying that cost attend more closely; under the positive-response condition derived below, the gap opens. A model-based minimum-distance calibration asks whether this mechanism can account jointly for the change in average forecasts and their dispersion without changing the groups' attention costs. The same choices must govern both outcomes. The benchmark reproduces much of the widening but understates the gap levels; individual cost heterogeneity substantially improves joint fit. This connects the mechanism to observed magnitudes and reveals which restrictions matter, while leaving absolute attention imperfectly determined.

Wealth and education stratify equity participation ([Haliassos and Bertaut, 1995](#); [Vissing-Jorgensen, 2003](#); [Guiso and Sodini, 2013](#)), so an informational advantage attached to it falls unevenly across households. The gap I document is wider for lower-income households and for

women. I condition on observed ownership and do not model the participation margin, so this describes where the gap sits rather than showing that ownership causes it.

The paper proceeds as follows. Section 2.2 reviews the related work; Section 2.3 develops the model; Section 2.4 and Section 2.5 describe the data and present the main estimates; Section 2.6 through Section 2.8 turn to mechanism, robustness, and quantitative calibration; Section 2.9 concludes.

2.2 Related Literature

Households disagree about future inflation to an extent that full-information rational expectations cannot accommodate, and that disagreement is neither random noise nor a fixed feature of the cross-section. Instead, it is something deeply rooted into their socio-economic and demographic characteristics (Coibion and Gorodnichenko, 2015, 2012; Mankiw et al., 2003). The dispersion of household forecasts is large, persistent and systematically related to observable characteristics, and it widens along with higher volatility in realized inflation (Mankiw et al., 2003; Patton and Timmermann, 2010). A large body of survey-based work has progressively emerged treating these subjective expectations as core economic objects: they can be measured (Manski, 2004; Engelberg et al., 2009), they differ sharply across households and they shape economic decisions, chiefly consumption and saving behavior (Coibion et al., 2018; Weber et al., 2022; Bachmann et al., 2015). Households appear to act on these beliefs: experimental and survey evidence shows that the inflation people expect feeds into their spending and portfolio choices (Armantier et al., 2015; Roth and Wohlfart, 2020). The divergence identified by several papers is deep enough that even when households and professionals are handed an identical information set, their beliefs about the direction and size of macroeconomic shocks remains widely dispersed (Andre et al., 2022). Observing the degree to which disagreement is systematic and responsive to conditions calls for a structural investigation of where it originates, what places a given household high or low in the distribution to begin with, and what moves it as inflation rises.

A first explanation for the positioning in this distribution identifies the source of disagreement in what households personally experience, rather than in what official statistics report (Malmendier

and Nagel, 2016; D’Acunto et al., 2021a; Cavallo et al., 2017). Individuals overweight the inflation they have lived through, so that lifetime experience predicts expectations and younger households, having shorter memories, update more aggressively (Malmendier and Nagel, 2016). At higher frequency, they anchor on the prices they encounter most often, the price changes in their grocery basket, weighted by how frequently goods are purchased rather than by expenditure share, and with increases having greater impact on memories than decreases (D’Acunto et al., 2021a). Reliance on personally experienced prices appears to persist even when respondents are shown official aggregate figures, pointing to the presence of genuine information frictions (Cavallo et al., 2017). Because the anchor is what a household sees rather than what it earns, the resulting bias is largest where price exposure is most salient: women, who historically perform most household grocery shopping, hold systematically higher expectations (D’Acunto et al., 2021b), households extrapolate from their own recent experience when forming aggregate forecasts (Kuchler and Zafar, 2019; Bordalo et al., 2020), and the belief gap across socioeconomic groups itself widens and narrows over the cycle (Das et al., 2020). Experienced prices, however, are only one of the inputs a household can draw on; another key one is the aggregate signal carried in macroeconomic news. Which of the two a household relies on more is itself a choice about how to allocate scarce attention, rather than a fixed trait.

Why households differ in how heavily they lean on experienced prices rather than aggregate signals is, in turn, a question about attention. Across the rational-inattention and behavioural-inattention traditions alike, attention is a costly and endogenous choice that responds to the information environment (Sims, 2003; Mankiw and Reis, 2002; Gabaix, 2014). Agents with finite processing capacity optimally direct attention toward whatever is most variable or most payoff-relevant, so the same household attends more to inflation exactly when inflation matters more (Sims, 2003; Maćkowiak and Wiederholt, 2009). Sticky-information and inattentive-consumer models make the point dynamically: households update infrequently, and those facing lower information costs update more often and forecast more accurately (Mankiw and Reis, 2002; Reis, 2006). Households moreover absorb macroeconomic news indirectly, through media reports of professional forecasters’ views, so the diffusion of information is itself a margin of attention (Carroll, 2003; Lamla and Lein, 2014); professional forecasters themselves display inattention to

public information, updating in ways consistent with sticky-information dynamics ([Andrade and Le Bihan, 2013](#)); and when attention is scarce, agents reason with sparse, simplified models that retain only first-order variables ([Gabaix, 2014](#); [Maćkowiak et al., 2023](#)). If attention is chosen in this way, it tracks the information environment a household happens to inhabit and that environment is not common to all. Some households move through daily life being in closer contact with macroeconomic information than others, which raises the question of what observable traits marks them out.

A smaller and more recent literature focuses precisely on identifying such traits, showing that equity-market participation in particular places households in a richer macroeconomic information environment ([Ahn and Xie, 2024](#); [Peress, 2014](#); [Jeong et al., 2025](#)). Most directly, [Ahn and Xie \(2024\)](#) document on the Michigan survey that stockholders forecast inflation more accurately, disagree less, and revise more responsively to news than non-stockholders, evidence that participation lowers the effective cost of attending to macroeconomic conditions. On the supply side, [Peress \(2014\)](#) show that the financial media causally diffuse macro-relevant information to market participants: when newspapers strike, trading volume and the dispersion of returns both fall. [Jeong et al. \(2025\)](#) formalize the link in a Gabaix-style attention model in which belief revisions to policy news concentrate among attentive, high-financial-stakes households, a first-order condition structurally isomorphic to the one developed below. This reading sits within a longer household-finance literature on why so few households hold equities and what separates participants from non-participants ([Haliassos and Bertaut, 1995](#); [Mankiw and Zeldes, 1991](#); [Vissing-Jorgensen, 2003](#); [Guiso and Sodini, 2013](#)), with financial literacy playing a first-order role in determining who holds stocks ([van Rooij et al., 2011](#); [Lusardi and Mitchell, 2014](#)) and the resulting investment quality differing substantially across the population ([Calvet et al., 2007](#)). That literature reads non-participation mainly as a wealth channel: households that stay out forgo the equity premium, and the resulting shortfall compounds over a working life ([Calvet et al., 2007](#)). The evidence below points to a second channel running alongside it. Participation also carries information about aggregate prices, so the households shut out of the asset are the same ones left to read inflation off their own shopping, and unequal access to the asset brings unequal access to the signal with it. Alongside theories in which attention is allocated toward

higher-stakes positions (Kacperczyk et al., 2016), direct measures confirm that investors attend to their portfolios selectively (Da et al., 2011; Sicherman et al., 2016). Read together, these strands describe a household whose place in the distribution of inflation beliefs is set at once by the prices it encounters, the attention it can spare, and the macroeconomic information its financial life brings within reach.

What these strands of literature have in common is that they read belief differences as standing feature. A prolonged inflation episode unsettles that reading, and poses a question none of them is built to answer: not how far apart two groups of households stand on average, but whether and why that distance changes when a shock endures. Placing an observable financial trait at the center, the paper takes the duration of an inflation episode as its source of variation and asks how the structure of disagreement responds as the episode lengthens. The remainder of the paper develops the two-signal attention model in Section 2.3, describes the Michigan survey data in Section 2.4, and then turns to the reduced-form, distributional, and quantitative assessment of the model.

2.3 A Model of Attention, Bundling, and Bias Amplification

2.3.1 Setup: signals and two household groups

A household forms a point forecast of π : the realised rate of headline consumer-price inflation over the following twelve months, expressed in percentage points and the same object the survey question puts to respondents. The forecast is built from two noisy signals. A *financial* signal summarises macro-financial news. The theoretical model assumes that its mean error is zero; attending to it costs attention. Unbiasedness is an assumption about the signal's mean, not a requirement that every realised forecast error or finite-episode average be zero. The calibration in Section 2.8 distinguishes this assumption from the realised errors of the observed series used to proxy the signal.

An *experienced-price* signal summarises the prices a household encounters and recalls in daily life. It costs nothing, since a household meets those prices anyway, but its relation to headline inflation is not automatic. Frequent purchase can give some prices more influence

than their expenditure share would imply, but frequency weighting alone does not guarantee an upward bias: the direction depends on which prices rise more quickly. In the present episode, the illustrative food-and-energy inflation proxy exceeds core inflation by 10.16 percentage points on average in 2021–22 and falls 3.05 points below it in 2023–November 2025. Evidence of frequency weighting and stronger responses to positive than negative grocery-price changes (D’Acunto et al., 2021a) motivates maintaining an upward experienced-signal bias $b_\tau > 0$ in the model, without identifying b_τ or a salience process in these data.

$$s_i^F = \pi + \varepsilon_i^F, \quad \varepsilon_i^F \sim \mathcal{N}(0, \sigma_F^2); \quad s_i^E = \pi + b_\tau + \varepsilon_i^E, \quad \varepsilon_i^E \sim \mathcal{N}(0, \sigma_E^2), \quad (2.3.1)$$

The parameters σ_F^2 and σ_E^2 are the noise variances of the two signals, measured in squared percentage points: each records how erratic that source is as a guide to π , independently of whether it is centred on the right value. The term $b_\tau > 0$ is the experienced-price bias at date τ , the systematic amount in percentage points by which the household’s impression of prices runs above the headline rate. It is a property of the signal, not a quantity the household nets out: as Section 2.3.2 sets out, b_τ enters the stakes against which attention is allocated, but the forecast rule carries no correction for it. The household combines the two signals with an attention weight $m_i \in [0, 1]$ on the financial signal,

$$E_i = m_i s_i^F + (1 - m_i) s_i^E = \pi + (1 - m_i) b_\tau + [m_i \varepsilon_i^F + (1 - m_i) \varepsilon_i^E], \quad (2.3.2)$$

so that a household attending more to financial news carries less of the experienced-price bias into its forecast.

Attending to the financial signal is costly: household i pays $\frac{\kappa_i}{2} m_i^2$ in utility to sustain attention m_i , where $\kappa_i > 0$ indexes how expensive attention is for that household. The quadratic form says that attention becomes harder at the margin: glancing at a headline is cheap, sustained monitoring is not, so marginal cost $\kappa_i m_i$ rises linearly in attention rather than staying flat. Households belong to one of two groups, stockholders and non-stockholders, $g \in \{S, N\}$, and κ_g is the group’s

effective attention cost rather than the average of what its members pay.² Stock ownership enters through a single channel: monitoring a portfolio bundles exposure to macro-financial information and so lowers the marginal cost of attending to it. Formally, stockholders receive a *bundling discount* $\delta > 0$ on their attention cost:

$$\kappa_S = \kappa_N - \delta, \quad \delta > 0, \quad (2.3.3)$$

This discount is grounded in the principle, common to the rational-inattention and behavioural-inattention traditions, that attention costs are endogenous to the information environment (Sims, 2003; Mankiw and Reis, 2002; Gabaix, 2014), and in direct evidence that stockholders attend more closely to macroeconomic conditions (Ahn and Xie, 2024; Peress, 2014). Stockholder status is taken as given throughout: the model describes how each group forms expectations, not who comes to hold stocks.

2.3.2 Optimal attention and the bundling discount δ

Each household chooses its attention weight to minimise expected squared forecast error plus the cost of attention,

$$\min_m (1 - m)^2 \Sigma_{E,\tau}^2 + m^2 \sigma_F^2 + \frac{\kappa_g}{2} m^2, \quad (2.3.4)$$

where $\Sigma_{E,\tau}^2 = \sigma_E^2 + b_\tau^2$ is the mean squared error of the experienced-price signal, decomposed as usual into variance and squared bias. The two are not conflated: a signal that is biased but tightly distributed can be a better guide than an unbiased but dispersed one. If inflation is 2 percent, a household whose impressions run 0.1 points high every month faces $\Sigma_E^2 = 0.01$, while one whose impressions are correct on average but alternate between 0 and 4 percent faces $\Sigma_E^2 = 4$.

What the forecasting problem responds to is the sum, not either component alone. The first-order

²Attention responds non-linearly to its cost, so the group's average attention is not the attention of a household at the group's average cost. In a group split evenly between costs of 1 and 99, average attention is 0.49 at the variances used later, while a household at the average cost of 50 would attend 0.18. I therefore define κ_g by $m^*(\kappa_g) = \mathbb{E}[m^*(\kappa_i) \mid g]$. The two definitions coincide when attention is common within a group, which is how the model is read throughout; households may differ in κ_i without disturbing anything below, but nothing here identifies that variation.

condition gives the optimal weight

$$m_{g,\tau}^* = \frac{\Sigma_{E,\tau}^2}{\sigma_F^2 + \Sigma_{E,\tau}^2 + \kappa_g/2}. \quad (2.3.5)$$

No constraint has to be imposed for this to be a well-defined attention weight. All three terms are non-negative, so the numerator is non-negative and never exceeds the denominator, and $m_{g,\tau}^*$ therefore lies in $[0, 1]$ by construction. The bounds bind only in the limits: the weight goes to zero as attention becomes infinitely costly, and to one as the experienced-price signal becomes infinitely poor.

The household in (2.3.4) is the behaviourally inattentive agent of [Gabaix \(2014\)](#). It allocates attention against the true stakes, evaluating the loss at the full second moment $\Sigma_{E,\tau}^2$ with the squared bias included, and it then acts under a simplified default model in which the correction does not appear: the forecast rule (2.3.2) carries no intercept, and b_τ is never netted out. $\Sigma_{E,\tau}^2$ is Gabaix's measure of stakes, the uncentred second moment of the variable attended to, and the household's attention to its own bias sits at a corner of zero, the counterpart of the nominal illusion in which a household that should subtract inflation from a nominal price change subtracts only part of it.

The alternative household is easy to characterise. Optimising over an intercept α_i in (2.3.2) as well as over m_i gives $\alpha^* = -(1 - m)b_\tau$: the bias leaves the forecast, leaves the loss, and leaves the attention weight, and the gap in (2.3.6) is zero at every date. The no-debiasing assumption is therefore a substantive behavioural restriction. Table 1 documents positive average forecast errors before 2021, but realised errors alone do not show that households knew their bias and failed to correct it. They can also reflect unanticipated aggregate shocks and other departures from the model.

Financial attention rises when experienced prices become a worse guide to inflation and falls when attention is more costly. The bundling discount (2.3.3) therefore gives stockholders a higher financial weight, $m_{S,\tau}^* > m_{N,\tau}^*$. Both groups can pay positive attention.

In the baseline model, taking group means in (2.3.2) gives

$$\mu_{g,\tau} = \pi + (1 - m_{g,\tau}^*)b_\tau, \quad \Delta_\tau \equiv \mu_{N,\tau} - \mu_{S,\tau} = (m_{S,\tau}^* - m_{N,\tau}^*)b_\tau. \quad (2.3.6)$$

The economic point is simple: stockholders rely less on the same upward-biased price experience, so their inflation expectations are lower. The size of the gap depends on both the bias in that experience and the difference in financial attention. A larger bias makes an existing attention difference matter more; a larger attention difference reduces stockholders' reliance on experienced prices relative to non-stockholders.

The gap need not keep widening. As the experienced signal becomes more biased, both groups have a stronger incentive to use financial information. With finite attention costs, sufficiently large bias eventually brings both weights close enough to one for the gap to narrow towards zero. This is a model implication; the empirical analysis does not establish where that convergence would occur. Appendix 2.A gives the derivation and explains how the theoretical bias relates to group means.

2.3.3 Duration and accumulated price experience

Persistent price pressure can influence expectations through the experience households retain from earlier months. To represent this accumulation, let

$$S_\tau = \rho S_{\tau-1} + \psi(p_\tau), \quad 0 < \rho \leq 1, \quad \psi(p_\tau) \geq 0. \quad (2.3.7)$$

Here S_τ is accumulated salient price experience, ρ is the fraction retained each month, and $\psi(p_\tau)$ is the contribution of current price pressure.

For constant pressure p over D months, starting from $S_0 = 0$,

$$S_D = \begin{cases} \psi(p) \frac{1 - \rho^D}{1 - \rho}, & \rho < 1, \\ D\psi(p), & \rho = 1. \end{cases} \quad (2.3.8)$$

When $\psi(p) > 0$, a longer episode therefore builds a larger stock even at unchanged monthly intensity. If $\rho < 1$, this accumulation gradually levels off. Greater intensity also increases the stock when ψ is increasing: both the duration and the intensity of price pressure matter.

Let accumulated price experience affect the experienced signal through

$$b_\tau = b_0 + \gamma S_\tau, \quad \gamma > 0. \quad (2.3.9)$$

Here b_0 is the baseline upward bias, while γS_τ is the additional bias associated with accumulated experience. The parameter γ measures how strongly that experience affects the signal: one additional unit of S_τ raises its mean error by γ percentage points. This linear relationship is a behavioural assumption. It describes the bias in the experienced signal; the household's final forecast also depends on the weight it gives financial information. For the duration comparison, both groups face the same mean signals and share this response, so their forecasts differ through attention.

This is the same b_τ as in (2.3.1). Since the financial signal has zero mean error in the model, the difference between the two signals' means is simply b_τ . Holding b_0 , noise variances and attention costs fixed, the assumed salience relationship therefore gives

$$\frac{\partial b_\tau}{\partial S_\tau} = \gamma.$$

No additional signal-bias parameter is needed for this duration comparison.

Write $A_\tau = m_{S,\tau}^* - m_{N,\tau}^*$ for the difference in the groups' weights on financial information. The expectations gap, defined as non-stockholder minus stockholder expectations, and its response to accumulated experience are

$$\begin{aligned} \Delta_\tau &= A_\tau b_\tau, \\ \frac{\partial \Delta_\tau}{\partial S_\tau} &= \gamma \left[A_\tau + b_\tau \frac{\partial A_\tau}{\partial b_\tau} \right]. \end{aligned} \quad (2.3.10)$$

For $b_\tau > 0$ and $A_\tau > 0$, the first term widens the gap: a larger difference between the signals makes the existing attention difference matter more. The second term captures changes in relative attention. It reinforces widening if stockholders' attention advantage grows and offsets it if non-stockholders catch up.

Duration therefore widens the gap provided convergence in attention does not offset the growing difference between the signals. Non-stockholders can pay positive and increasing attention throughout. Appendix 2.A, equation (2.A.7), derives this response from the attention

rule.

The mechanism does not imply indefinitely increasing disagreement. Accumulated experience can decay when $\rho < 1$ and new price pressure weakens; convergence in attention can also narrow the gap. The salience process is illustrative and is not estimated here, so this is a conditional theoretical mechanism rather than a separately identified causal effect of duration. The distinction between forecast-target errors and empirical proxy residuals is discussed in Section 2.8.

2.3.4 Testable predictions

The model yields three conditional predictions about the timing, distribution and household heterogeneity of the expectations gap. Write $A_\tau = m_{S,\tau}^* - m_{N,\tau}^* > 0$ for the financial-attention difference and define Δ_τ as non-stockholder minus stockholder expectations.

Prediction 1 (Duration widening). Under (2.3.7), sustained price pressure with $\psi(p) > 0$ builds accumulated experience S_τ even at unchanged monthly intensity. With a common salience response $\gamma > 0$, this increases the experienced signal's mean bias b_τ , which is also the difference between the two signals' means. The group gap is

$$\Delta_\tau = A_\tau b_\tau.$$

For $b_\tau > 0$, accumulated experience widens this gap provided the attention difference does not shrink fast enough to offset the growing signal wedge, as specified in (2.3.10). Persistent episodes can therefore generate widening over time even when both groups attend. The post-2021 timing and the comparison with the shorter 2008 price surge in Section 2.5 assess this pattern; they do not separately identify the causal effect of duration.

Prediction 2 (Tail compression). Greater financial attention reduces the weight $1 - m_{g,\tau}^*$ on experienced prices in (2.3.2), including unusually high individual price experiences. With attention common within each group, a common financial signal and a shared experienced-signal distribution, the gap at quantile q satisfies

$$Q_q(E_\tau^N) - Q_q(E_\tau^S) = A_\tau [Q_q(s_\tau^E) - s_\tau^F].$$

The gap increases across quantiles and is positive above the financial anchor: greater attention pulls high forecasts down further. Both groups can pay positive attention. Under Gaussian experienced-signal noise, the gradient is governed by $A_\tau \sigma_{E,\tau}$. An increase in this product after the surge predicts a more negative stockholder-minus-non-stockholder change at higher quantiles. The quantile estimates in Section 2.6 provide an empirical comparison with this conditional distributional prediction.

Prediction 3 (Salience sensitivity). Households differ in how strongly accumulated price experience affects their perceived signal. The parameter γ_i captures this sensitivity:

$$b_{i,\tau} = b_0 + \gamma_i S_\tau, \quad \frac{\partial b_{i,\tau}}{\partial \gamma_i} = S_\tau.$$

At $S_\tau > 0$, higher sensitivity produces a larger experienced-signal bias. Comparing otherwise similar households at the two group attention costs, this translates into a larger expectations gap under the positive-response condition in Prediction 1, holding other signal inputs fixed. Gender and income provide imperfect indicators of exposure; research linking grocery-shopping roles to inflation expectations motivates the gender comparison ([D’Acunto et al., 2021b](#)). The regressions in Section 2.6 examine these patterns jointly with education. They assess whether gender moderation persists after allowing socioeconomic moderation, without requiring an education null or identifying exposure separately from attention costs, financial knowledge and reporting behaviour.

2.4 Data

2.4.1 The Michigan Survey of Consumers

The data come from the University of Michigan Survey of Consumers, a nationally representative survey of U.S. households which has been conducted monthly since 1978. The survey is of rotational nature with a repeated cross-section: each month the survey interviews around five hundred households, a portion of whom are re-interviewed six months later. The central variable is the household’s quantitative one-year-ahead inflation expectation: the expected percent change

in prices over the next twelve months. Following standard practice I discard non-substantive responses and restrict the variable to the range $[-5, 50]$ percent to remove implausible outliers. All means, regressions, and distributional statistics use the survey's sampling weights, except where an estimator does not admit them, as noted in place. The stock-ownership question is asked on a continuous monthly basis only from 1997 onward, so the estimation sample runs essentially from 1997 through 2025, spanning the tranquil pre-pandemic decades, the 2008 financial crisis, and the 2021–2025 inflation surge. The item was also fielded in 1990, and those months are retained; they account for about one percent of the estimation sample.

2.4.2 Variable construction and stockholder status

A household's classification as a stockholder is based on the survey's investment item, which asks whether the household holds any investments in the stock market: I set $Stock_i = 1$ for respondents who answer yes and 0 for those who answer no, dropping the small number who do not respond. The ownership question is put to every respondent, whereas the dollar amount held is asked only of those who own and answered by a subset of them, which is why the treatment variable is binary. The size of the position enters separately: Table 7 uses the reported amount directly and finds the effect rising monotonically across quartiles of holdings, on forecast accuracy as well as on the level, so the binary treatment is not concealing a flat response at the intensive margin. The treatment timing follows the surge, with $Post_t = 1$ from January 2021 onward and 0 before. The coefficient of interest throughout is on the interaction $Stock_i \times Post_t$, which measures how stockholders' expectations moved relative to non-stockholders' as the surge took hold.

The remaining covariates are built from the survey's standard demographic and socioeconomic items. Two demographic indicators identify respondents under 45 and women. Two socioeconomic indicators identify college graduates, those holding at least a bachelor's degree, and high-income households, defined as the top two quintiles of the survey's income distribution. A homeownership indicator is constructed in the same binary fashion for use in the housing robustness check.

Two further items are constructed for later use. A specific question records the dollar value a stockholder reports holding in the market. I discard the survey's sentinel non-response codes

and, where the reported amount is positive, take natural logs. A separate item, available from 2007, records the household's one-year-ahead expectation for home prices and is cleaned in the same way as the inflation expectation.

2.4.3 Summary statistics

Table 1 reports survey-weighted means for the pre-2021 sample, split by stock ownership. Before the surge the two groups already differed in expectations, but only modestly: stockholders expected inflation of 3.30 percent on average against 4.07 percent for non-stockholders, a gap of 0.76 percentage points that had been roughly stable for two decades. Stockholders were also more educated and higher-income, as one would expect (55 against 22 percent college graduates, and 54 against 14 percent in the top income quintiles), while the two groups were similar in age and gender composition. The pre-2021 gap is not itself an artefact of that composition: Panel B shows it present within every demographic and socioeconomic category, from 0.37 points among high-income households to 0.79 among those under 45. These compositional differences are precisely why the design conditions on them: the question is not whether stockholders differ from non-stockholders, but whether the gap in expectations *widened* as the surge persisted, beyond what these stable traits can explain.

Table 1: Summary statistics, pre-2021 period

	Stockholders	Non-stockholders	Difference
	mean (standard deviation)		
<i>Panel A. Household characteristics</i>			
One-year inflation expectation (%)	3.30 (3.44)	4.07 (4.22)	−0.76
<i>memo:</i> realised inflation (%)	2.28 (1.43)	2.27 (1.33)	+0.01
Age below 45 (share)	0.39 (0.49)	0.41 (0.49)	−0.02
Female (share)	0.49 (0.50)	0.57 (0.50)	−0.08
College graduate (share)	0.55 (0.50)	0.22 (0.42)	+0.33
High income (share)	0.54 (0.50)	0.14 (0.34)	+0.40
<i>Panel B. One-year inflation expectation (%), by category</i>			
College graduate	3.04 (3.11)	3.62 (3.78)	−0.58
No college degree	3.64 (3.78)	4.21 (4.34)	−0.56
High income	3.02 (3.16)	3.39 (3.72)	−0.37
Lower income	3.64 (3.71)	4.19 (4.29)	−0.54
Women	3.65 (3.77)	4.39 (4.46)	−0.73
Men	2.99 (3.08)	3.69 (3.90)	−0.70
Age below 45	3.12 (3.38)	3.90 (4.15)	−0.79
Age 45 and over	3.42 (3.47)	4.19 (4.27)	−0.76
Observations	90,636	54,448	

Notes: Pre-2021 sample (survey months before January 2021) with non-missing stock-ownership status. Entries are survey-weighted means; the final column is the stockholder-minus-non-stockholder difference in means. Panel B reports the same expectation variable within each demographic and socioeconomic category, so the categories partition the sample rather than nesting inside one another. The observation counts are the pre-2021 sample with non-missing ownership status. The expectation and realised-inflation rows use cleaned forecasts, observed twelve-month outcomes and positive survey weights: 82,395 stockholders and 43,201 non-stockholders. The realised-inflation reference is averaged over each group's own respondent-months using the same weights as its forecast mean. Average signed forecast errors, calculated before rounding, are 1.03 points for stockholders and 1.80 for non-stockholders.

2.5 Empirical Strategy and Main Results

This section will introduce the main Difference-in-differences specification, detailing the features of the econometric techniques adopted. The main results will be presented and analysed next, together with some of the core robustness checks which are specific to the Difference-in-differences approach.

2.5.1 Difference-in-differences specification

The estimating equation compares the change in stockholders' inflation expectations across the surge to the contemporaneous change for non-stockholders:

$$E_{i,t} = \beta (Stock_i \times Post_t) + \gamma Stock_i + X'_{i,t}\theta + \lambda_t + \varepsilon_{i,t}, \quad (2.5.1)$$

where $E_{i,t}$ is the one-year-ahead expectation, λ_t are survey-year fixed effects that absorb the $Post_t$ main effect and any aggregate movement in expectations, $X_{i,t}$ is a vector of controls, and standard errors are clustered by survey month. The parameter of interest is β , the coefficient on $Stock_i \times Post_t$ which measures how much stockholders' expectations rose relative to non-stockholders' once the surge took hold. The design conditions on reported stock ownership in a repeated cross-section, with a common post-period cutoff. It does not require ownership to remain fixed for every household. The heterogeneous-timing estimators developed for staggered adoption ([Callaway and Sant'Anna, 2021](#); [de Chaisemartin and D'Haultfœuille, 2020](#)) address a problem this design does not have, and are not used here.

I report four specifications that add controls in layers. The *baseline* includes only the stockholder indicator and year fixed effects. *+Demographics* adds the age and gender indicators; *+SES* adds the college-graduate and high-income indicators and the college-graduate interaction with $Post$. Each specification also includes the missing-category indicators for its controls. The headline *+Demog. $\times$ Post* specification interacts every demographic and socioeconomic control with $Post_t$, so that β is identified only off the stockholder-specific change across the surge, net of any change in expectations common to younger, female, college-educated, or higher-income households over the same period. This is the most demanding of the four and the one I treat as the headline. A fifth specification adding a homeownership interaction is reported as a robustness check in Section 2.7.

2.5.2 Baseline estimates

Table 2 reports the four specifications. The baseline estimate is -2.477 percentage points. Adding demographic controls gives -2.569 ; adding socioeconomic controls and the college-

by-post interaction gives -2.038 . The headline specification, which allows every observed and missing demographic category its own post-period shift, gives -1.784 (SE 0.198) on 166,125 observations. About 72 percent of the baseline coefficient’s magnitude remains, and all four estimates are significant at the 1 percent level. The headline is the comparison carried through the paper; the homeownership extension is reported separately in Section 2.7.

Table 2: Stock ownership and inflation expectations: difference-in-differences

	Baseline	+Demog.	+SES	+Demog. $\times$ Post
Stock $\times$ Post	-2.477^{***} (0.220)	-2.569^{***} (0.230)	-2.038^{***} (0.220)	-1.784^{***} (0.198)
Survey-year FE	Yes	Yes	Yes	Yes
Demographic controls	No	Yes	Yes	Yes
Socioeconomic controls	No	No	Yes	Yes
Controls $\times$ Post	No	No	College only	All
Observations	166,125			

Notes: Dependent variable: cleaned one-year-ahead inflation expectation. Each column reports the coefficient on *Stock $\times$ Post* from a separate survey-weighted regression; standard errors clustered by survey month in parentheses. All columns include survey-year fixed effects. The headline specification is the +Demog. $\times$ Post column. $^{***}p < 0.01$.

2.5.3 Event study

To see *when* the gap opened, I replace the single *Post* interaction with a full set of *Stock_i $\times$ year* interactions, normalised to the last pre-surge year (2019). Figure 2 plots the resulting coefficients. Through 2020 they are small and hover near zero. The largest pre-period deviation, in 2010, is 0.49 percentage points. A joint test that the 2012–2018 leads are zero does not reject ($p = 0.156$), but that is a narrow window: testing every displayed untreated year, 2007–2018 and 2020, rejects ($p < 0.001$), and the rejection is driven by 2007, 2010 and 2016. Both tests are stated for the 2019 reference year; a different reference is a different null hypothesis. Pre-treatment differences are therefore small in magnitude but not jointly zero, which is a reason to lean on the sensitivity analysis below rather than to treat parallel trends as established. The gap then opens only after the surge and *widens as it persists*: the 2021 coefficient is still statistically indistinguishable from zero, but it falls to -1.32 in 2022, -2.32 in 2023, and -2.38 in 2024. The effect is thus not an instantaneous response to the onset of inflation but a divergence that builds over the duration of

the episode, the reduced-form signature of Prediction 1.

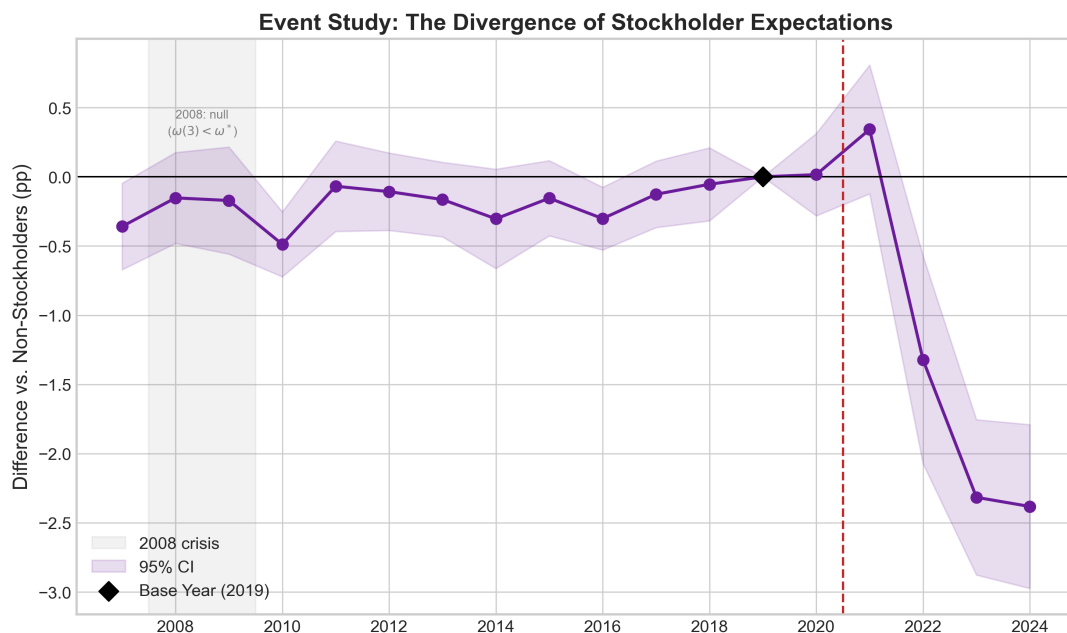


Figure 2: Event study of the stockholder–non-stockholder expectations gap. Coefficients on *Stock* × year interactions from the difference-in-differences specification, normalised to 2019; bands are 95% confidence intervals from standard errors clustered by survey month.

2.5.4 The longer episode and the delayed widening of the gap

The comparison with 2008 provides an episode contrast. In one survey-weighted regression, I interact stock ownership with a 2008–2010 indicator and a post-January-2021 indicator, allowing demographic and missingness categories their own shifts in both windows. The earlier interaction is -0.025 (SE 0.088), compared with -1.785 (SE 0.198) after 2021. Their difference is 1.760 (SE 0.213); allowing serial dependence with monthly-score HAC(12) increases the SE of the difference to 0.387, while its interval still excludes zero. Defining the earlier window as 2008 alone, 2008–2009 or 2007–2009 preserves the unequal episode responses.

The later episode was substantially longer. Headline inflation remained above four percent for five consecutive months in 2008 and for twenty-six months, April 2021–May 2023, in the later episode. Thresholds of three and five percent give the same ordering: thirteen versus thirty-eight months, and two versus twenty-one. Peak headline inflation was also higher after 2021, 8.98 percent against 5.50 percent, so the episodes are not matched shocks and equality of

the episode coefficients is not a test of a duration effect holding magnitude constant. The peak nonetheless overstates the difference in the pressure households were exposed to. Within the two spells, mean headline inflation was 6.64 percent after 2021 against 4.96 percent in 2008, and the food-and-energy signal that anchors experienced-price beliefs averaged 14.02 percent in the later spell against 14.97 percent in the earlier one, marginally the higher of the two. The episodes therefore differ far more in how long elevated prices persisted, by a factor of five, than in how intense they were while they persisted.

The timing within the later episode provides additional evidence. Allowing the same controls their own changes in 2021–22 and from 2023 onward, I directly test whether the group gap became larger after the inflation peak. It did: the adjusted non-stockholder-minus-stockholder gap widened by 1.41 percentage points in 2023 relative to 2021–22, with HAC(12) SE 0.44. The result precedes the survey’s April 2024 transition toward web interviewing and also holds when the sample is restricted to phone interviews (Table 3).

Table 3: Widening of the expectations gap after 2021–22

Later period / sample	Widening (pp)	Month-cluster SE	HAC(12) SE
2023	1.406	0.367	0.442
2023–2024	1.554	0.333	0.462
2023–November 2025	1.759	0.325	0.471
Phone interviews only	1.455	0.331	0.442

Notes: Positive values denote an increase in the non-stockholder minus stockholder expectations gap relative to 2021–22. Survey-weighted WLS with year fixed effects and sex, age, education and income indicators, explicit item-nonresponse categories, and interactions of all these categories with both phases. Sample sizes are 145,565; 154,815; 166,125; and 146,801 respectively. The phone-only later phase ends in June 2024. HAC uses Bartlett-weighted autocovariances of the monthly estimating equations through lag 12. Replacing year effects with month-year effects gives 1.391, 1.564, 1.714 and 1.443; the HAC intervals continue to exclude zero. The small number of post years limits the precision of inference about episode-level persistence.

The pattern is consistent with accumulated price experience: disagreement develops over a sustained episode and remains large during disinflation. It does not uniquely identify that mechanism. A delayed response to the size of the shock, heterogeneous adaptive learning, changes in anchors or other episode-specific changes can also generate persistence. The contribution is the documented evolution of the group gap and its consistency with the proposed duration mechanism; the design does not isolate duration from those alternatives.

Two further placebo cutoffs (2015 and 2018), reported in Section 2.7, similarly produce no comparable widening.

Table 4: Both episodes in a single regression

	(1)	(2)	(3)
$Stock \times D^{2008}$	-0.145* (0.084)	-0.124 (0.084)	-0.025 (0.088)
$Stock \times D^{post}$	-2.494*** (0.220)	-2.559*** (0.230)	-1.785*** (0.198)
Difference	+2.349*** (0.232)	+2.435*** (0.242)	+1.760*** (0.213)
p -value, H_0 : equality	< 0.001	< 0.001	< 0.001
Demographic levels	No	Yes	Yes
Demographics $\times$ both windows	No	No	Yes
Year fixed effects	Yes	Yes	Yes
Observations	166,125	166,125	166,125

Notes: One survey-weighted regression per column on the full sample. D^{2008} equals one for 2008–2010 and D^{post} for months from January 2021; months outside both windows are the control period. Because both windows are unions of whole years, their main effects are absorbed by the year fixed effects. Column (3) interacts each demographic control with both windows, so the two episodes carry identical controls; its $Stock \times D^{post}$ coefficient closely matches the headline estimate of Table 2. “Difference” is the Wald contrast $Stock \times D^{2008} - Stock \times D^{post}$. Standard errors clustered by survey month in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

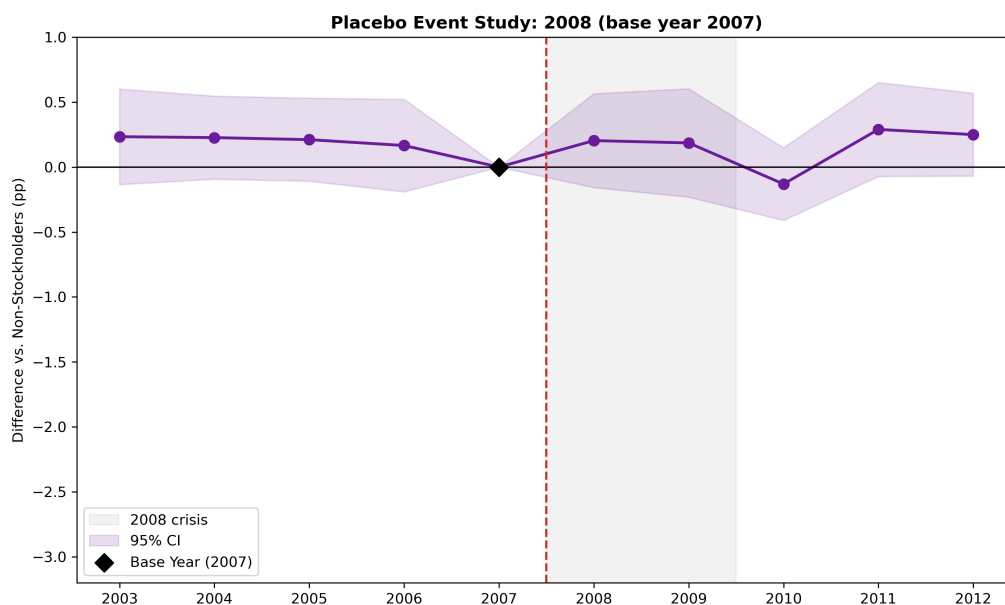


Figure 3: Event study around the 2008 commodity-price spike. Coefficients on $Stock \times Year$ relative to a 2007 base, estimated on 2003–2012 ($N = 50,574$) with the controls of the 2008 placebo. The vertical scale is that of Figure 2.

Current price changes and accumulated experience. The experienced-price proxy is the equally weighted average of the year-on-year food-and-beverage and energy inflation rates. This illustrative basket is not a household-specific purchase-frequency index. Its wedge over core inflation turns negative after 2022, even as the expectations gap remains large (Figure 4). The January-2021-rebased price-level index nonetheless remains 7.30 index points above the corresponding core index in November 2025. The latter is a cumulative relative-price difference, not a current inflation rate or a forecast bias. The contrast motivates a distinction between current price changes and their possible persistence in memory.

The evidence also limits a duration interpretation based directly on the positive salient-price wedge. The post-2021 positive-wedge spell lasts twenty-five months, while the pre-2021 data include a forty-six-month spell from November 2002 to August 2006. The previously tabulated twenty-nine-month spell runs from November 2009 to March 2012, rather than describing the 2008 spike itself. The unusually long headline-inflation episode and the later expectations gap therefore cannot be restated as a universal record duration for every salient-price measure.

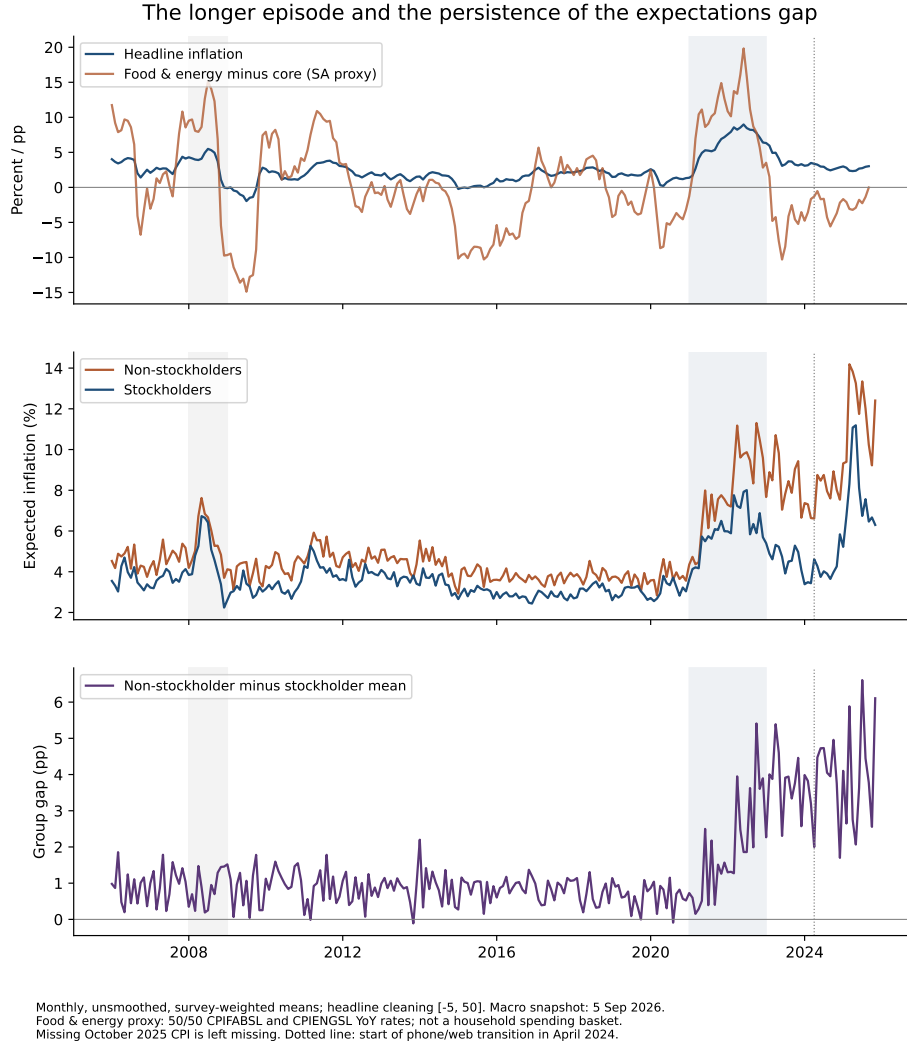


Figure 4: Inflation, salient-price pressure and the expectations gap

Sensitivity to the experienced-price horizon. I calculate the experienced-price proxy from annualised cumulative price changes over 12, 24, 36, 48 and 60 calendar months. The comparison holds available months fixed across horizons and leaves missing CPI observations unfilled. Under the headline survey cleaning and the seasonally adjusted food-and-beverage proxy, the twelve-month residual $\bar{E}_N - \pi^E$ is 1.23 percentage points before 2021, -7.16 in 2021–22 and 8.55 in 2023–November 2025. It cannot be represented as a uniformly positive, steadily increasing perceptual bias.

All five horizons give a larger pooled post-2021 residual than before 2021. Only the 36- and 60-month measures increase across all three phases; the 24-month early residual is negative and the 48-month residual falls slightly between the early and later phases. Alternative survey

cleaning and the NSA food series preserve that pass/fail pattern. These are measurement sensitivities, not a procedure for selecting the true memory horizon. Adopting a longer window changes the signal proxy and does not establish that the twelve-month sign reversal is an artifact.

Adaptive responses as an alternative. The delayed widening is also compatible with heterogeneous adaptive expectations. A rule such as $E_{g,t} = (1 - \lambda_g)E_{g,t-1} + \lambda_g Z_{g,t}$ can generate different adjustment paths when gains or information inputs differ between groups. The present design does not distinguish that account conclusively from accumulated salience and differential attention. In particular, the sign of a Mincer–Zarnowitz slope is not a test of this updating equation: it relates the forecast to subsequent realised inflation. With a common frozen data cutoff and HAC(12) inference, the post-2021 non-stockholder slope is -0.480 (SE 0.277 , $p = 0.083$), which offers limited evidence against zero and does not rule out adaptive responses.

2.5.5 Forecast accuracy and Mincer–Zarnowitz

The lower stockholder forecasts are accompanied by an improvement in accuracy. Using the headline controls, including missing-category indicators and their post interactions, stockholders’ absolute forecast errors fell by 1.381 percentage points more than non-stockholders’ (SE 0.181). Signed errors fell by 1.497 points more (SE 0.188), on the same 159,954 observations (Table 5). Both estimates use frozen CPI inputs on a complete monthly calendar without filling missing observations. They establish a relative accuracy improvement, not the particular information-processing channel responsible for it.³

The Mincer–Zarnowitz regression asks a different question: whether monthly group-mean forecasts track subsequent realised inflation. Over the 53 post-2021 months with complete inputs through June 2025, the non-stockholder slope is -0.480 (HAC(12) SE 0.277 , $p = 0.083$ against zero), while the stockholder slope is 0.067 (SE 0.236 , $p = 0.777$). These estimates offer limited evidence about the signs of the slopes. In particular, the negative non-stockholder point estimate does not identify bias amplification or reject adaptive expectations. The hypothesis that a slope

³Regressing forecast error on the forecast itself is not an efficiency test. Because the dependent variable is the forecast minus realised inflation, its coefficient is one minus the corresponding slope in a realised-inflation regression. With month-year fixed effects it equals one mechanically, since realised inflation is common to respondents in a month. The accuracy tables therefore report only signed and absolute forecast errors.

equals zero also differs from the joint calibration restriction of a zero intercept and unit slope.

Table 5: Stock ownership and forecast accuracy

	Absolute error	Signed bias
Stock $\times$ Post	−1.381*** (0.181)	−1.497*** (0.188)
Stockholder	−0.470*** (0.024)	−0.494*** (0.028)
Headline controls	Yes	Yes
Observations	159,954	

Notes: The dependent variable is the absolute forecast error (column 1) and the signed forecast error, i.e. expectation minus realised inflation (column 2). Both columns use the headline controls; standard errors clustered by survey month in parentheses. ** $p < 0.05$, *** $p < 0.01$.

2.6 Mechanism

This section is built on a comprehensive set of post-estimation tests which are aimed at unpacking the mechanisms behind the observed widening gap in inflation expectations between stockholders and non-stockholders.

2.6.1 Quantile regression

The headline coefficient describes an average. Prediction 2 instead asks whether the widening is concentrated among high forecasts. Survey-weighted quantile regressions ([Koenker and Bassett, 1978](#)), with the headline controls and year effects estimated jointly, give Stock $\times$ Post coefficients of 0.000, −0.714, −2.500 and −4.667 at the 25th, 50th, 75th and 90th percentiles. The corresponding survey-month bootstrap standard errors are 0.110, 0.231, 0.448 and 0.850, based on 399 joint replications. The 95 percent percentile interval includes zero at the lower quartile and excludes it at the median ([−1.000, −0.167]). Resampling whole years within the pre- and post-periods gives a wider median interval, [−1.000, 0.288]. The upper-quartile and 90th-percentile intervals exclude zero under both schemes (Figure 5).⁴

⁴Each replication resamples clusters and estimates all four quantiles using the joint weighted quantile objective. Year effects are included in that objective, not removed by an OLS regression beforehand. The year-cluster exercise allows dependence across months within a year, but its short post-period limits inference.

The contrast is about changes in conditional quantiles, rather than a fixed difference in the groups' forecast dispersions. Its upper-tail concentration survives the more conservative year-cluster comparison. The 2008 episode contrast concerns mean forecasts and does not separately test this tail prediction.

Winsorisation provides a related check using the same headline specification as the unmodified outcome. The coefficient moves from -1.784 to -1.030 (SE 0.144) under 1/99 winsorisation and to -0.349 (SE 0.093) under 5/95 winsorisation, reductions of 42 and 80 percent in magnitude. Excluding forecasts above 20 percent gives 0.046 (SE 0.108 , $p = 0.667$). High reported forecasts therefore account for a substantial part of the average widening, although a negative difference remains after winsorisation.

The model gives a conditional explanation for this gradient. Under the shared-signal and common-attention restrictions, financial attention pulls forecasts toward a common anchor, with a larger reduction for forecasts further above it. Section 2.8 states the mapping and distinguishes its implications from the moments used in calibration. The controlled quantile-regression gradient is consistent with this mechanism; it is not a numerical validation of the homogeneous calibration, and the unconditional mapping does not directly predict these conditional regression coefficients.

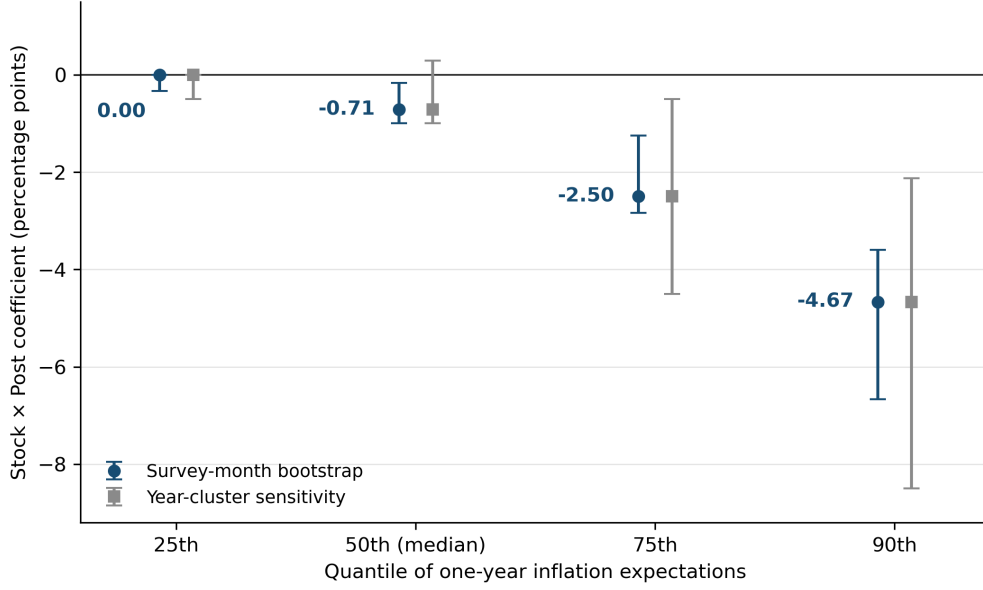


Figure 5: Quantile gradient of the stockholder–non-stockholder gap. Coefficients on $Stock \times Post$ from survey-weighted quantile regressions with headline controls, explicit missing categories and jointly estimated year effects ($N = 166,125$). Bars are 95 percent percentile intervals from 399 joint survey-month bootstrap draws and 399 year-cluster draws, the latter stratified by pre- and post-period.

2.6.2 Heterogeneity

Prediction 3 links heterogeneity in the expectations gap to differences in salience sensitivity, under the model’s positive-response condition. Triple-interaction regressions add $Stock \times Post \times X_i$ and the corresponding lower-order terms to the headline specification for a binary characteristic X_i (Table 6). The stockholder effect is -2.49 (SE 0.31) among women against -1.16 (SE 0.19) among men, and -1.95 (SE 0.23) among lower-income households against -0.94 (SE 0.32) among higher-income households. These are observable differences in the response of expectations; interpreting them as variation in γ_i requires additional assumptions about the proxies.

College graduates also show a smaller effect, -1.37 (SE 0.26) against -2.08 (SE 0.28) among non-graduates. The education interaction is $+0.71$ (SE 0.36, $p = 0.0493$), so the evidence does not support treating education as irrelevant. A regression including both education and income moderation cannot reject equality of their interactions (difference -0.34 , SE 0.55, $p = 0.54$), while rejecting that both are zero ($p = 0.023$). This indicates uncertainty about the difference between two estimated gradients; it does not establish that they are a single effect. The age

interaction is +0.55 (SE 0.31).

Gender moderation persists when education and income moderation enter jointly: the female interaction is -1.35 (SE 0.31, $p < 0.001$). It remains negative and significant across the two- and three-moderator specifications under all three missing-data policies.⁵ This conditional gender gradient is consistent with an exposure interpretation, which research on grocery-shopping roles motivates (D’Acunto et al., 2021b). Gender also captures other household differences. These regressions consequently do not isolate exposure from attention costs, financial knowledge or reporting behaviour, and do not rule out explanations involving those channels.

Table 6: Heterogeneity in the stockholder effect (triple interactions)

	Gender (female)	Income (high income)	Education (college)	Age (young)
Stock $\times$ Post $\times$ Char.	-1.329^{***} (0.322)	$+1.015^{**}$ (0.410)	$+0.710^{**}$ (0.361)	$+0.552^*$ (0.306)
Effect where Char. = 0	-1.160^{***} (0.192)	-1.951^{***} (0.232)	-2.079^{***} (0.276)	-2.005^{***} (0.234)
Effect where Char. = 1	-2.489^{***} (0.306)	-0.936^{***} (0.323)	-1.368^{***} (0.263)	-1.452^{***} (0.274)
Observations	166,125			

Notes: Each column adds the full set of two-way and three-way interactions of *Stock*, *Post*, and the demographic characteristic to the headline specification, so the remaining demographic controls and their *Post* interactions are included alongside the moderator; the row reports the triple interaction. A negative coefficient means the stockholder effect is larger (more negative) for that group. The two lower rows give the stockholder effect within each level of the characteristic, obtained as *Stock* $\times$ *Post* and as its sum with the triple interaction, with standard errors from the same covariance matrix. Missing-status indicators and their *Post* interactions are included, along with the interaction of stockholding and the moderator’s missing-status indicator; redundant columns are omitted. The education interaction has $p = 0.0493$; interpretation does not depend on classifying it around the five percent threshold. Survey-weighted estimates; standard errors clustered by survey month in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

2.6.3 Within-stockholder dose-response

The group-average effect also masks a gradient *within* stockholders. Sorting stockholders by the dollar amount they report holding in the market, the effect rises monotonically with the size

⁵In the three-moderator specification, the college and female interactions differ in signed value (1.97, SE 0.46, $p < 0.001$). They have opposite signs, however, and equality of their magnitudes in those directions is not rejected: the sum of the two coefficients is -0.72 (SE 0.48, $p = 0.133$). The signed comparison therefore does not establish that gender moderates the effect more strongly than education.

of the position. Relative to the smallest-holding quartile, expectations fall by 0.45, 0.89, and 1.63 percentage points more across the second, third, and top quartiles after the surge and, more tellingly, absolute forecast errors fall by 0.32, 0.48, and 1.00 points (Table 7); the continuous specification gives a log-holding interaction of -0.32 (SE 0.04). The accuracy gradient in the second column is the key result: larger holders do not merely report lower numbers, they forecast more accurately. A placebo that assigns a pseudo-surge in 2014 to the pre-2021 sample produces no gradient on any quartile, and the pattern survives saturating the regression with observables (about a tenth of the gradient attenuates).

I read this as descriptive enrichment of the mechanism, not a structural estimate. It shows that the group-average treatment effect summarises a distribution of responses that widens with stakes. A plausible reading is stakes-driven attention allocation in the spirit of [Sims \(2003\)](#) (households with more at risk attend more), but because the design conditions on stockholder status and observed holdings and cannot absorb unobserved wealth or preferences, I do not interpret the gradient as a structural attention parameter.

Table 7: Within-stockholder dose-response by holding size

INVAMT quartile $\times$ Post	Expectation	Absolute error
Q2	-0.452^{**} (0.218)	-0.320^* (0.178)
Q3	-0.887^{***} (0.169)	-0.483^{***} (0.133)
Q4 (largest)	-1.625^{***} (0.232)	-1.002^{***} (0.234)
Observations	92,980	88,687

Notes: Within-stockholder sample. Quartiles of reported INVAMT; the smallest-holding quartile (Q1) is the omitted reference. Each column is a separate survey-weighted regression with the headline controls; standard errors clustered by survey month in parentheses. The continuous analogue is $\log(\text{INVAMT}) \times \text{Post} = -0.321$ (SE 0.043). A placebo assigning a pseudo-post in 2014 yields no significant gradient. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

2.7 Robustness

The headline survives a battery of checks, collected in Table 8 and summarised here.

Additional placebo cutoffs. Neither the 2015 estimate (-0.098 , SE 0.072, $p = 0.176$) nor

the 2018 estimate (0.039, SE 0.081, $p = 0.635$) differs significantly from zero. The standalone 2008 placebo is also close to zero (0.006, SE 0.103). It uses the 2003–2010 sample and differs from the pooled episode regression in Table 4, whose fully interacted 2008–2010 coefficient is -0.025 (SE 0.088). These are separate specifications, both showing little mean widening around the earlier episode.

Sensitivity to parallel-trend violations. Pre-treatment differences are not jointly zero, and a placebo cannot rule out changing selection. The [Rambachan and Roth \(2023\)](#) relative-magnitude sensitivity analysis uses the equal-weight average of the 2021–2024 event-study coefficients relative to 2020. Its 95 percent interval excludes zero at $\bar{M} = 0.80$ ($[-2.90, -0.05]$) and includes zero at $\bar{M} = 0.85$ ($[-2.98, 0.03]$). Thus this average remains negative when the permitted post-period violation is up to about 0.8 times the largest pre-period violation. This is a sensitivity result for the annual event-study average, not the pooled headline coefficient or a test excluding selection.

Time fixed effects. Replacing year effects with month-year effects, holding the headline controls and sample fixed, gives -1.777 (SE 0.199) against -1.784 (SE 0.198). The difference is 0.006 percentage points. The headline result therefore changes little when common shocks are absorbed at the survey’s monthly frequency.

Alternative treatment timing. Moving the post-period start to March 2021, July 2021, or January 2022 leaves the estimate in the same range ($-1.71, -1.63, -2.34$); the effect is not an artifact of the exact cutoff.

Sample and composition. Restricting to phone respondents (-1.892) and reweighting cells to hold demographic composition fixed (-2.402) both preserve the effect, as does adding a homeownership interaction (-1.618). These checks preserve the widening under alternative sample and observed-composition adjustments.

Compositional change in the stockholder group. Reported ownership increases from about 60 to 68 percent across the cutoff, so group membership can change. Demographic controls and reweighting address observed composition, but they do not establish that selection on unobserved characteristics is absent. The holdings gradient in Section 2.6 compares larger holders with Q1, the omitted smallest-holding quartile. Its zero is a normalisation, not an estimated zero effect

for small holders relative to non-stockholders. Holdings also do not establish entry dates or portfolio engagement. This gradient therefore cannot show that new entrants left the widening unchanged or made the headline estimate a lower bound.

Partisanship. Since partisan alignment tracks the party in power, I add a triple interaction with an out-of-power indicator on the full-SES sample; the stockholder effect is unchanged (-2.075), so it is not a relabelled partisan response.

Specificity to inflation. The home-price absolute-error interaction is -0.181 (SE 0.133 , $p = 0.172$). There is no statistically detectable relative improvement in home-price forecast accuracy. This comparison does not establish that the consumer-price result is caused by a specific information channel or rule out differences in general forecasting ability.

Within-household panel. The rotating-panel regression gives -0.124 (HC1 SE 0.292 , $p = 0.672$) on 51,664 paired changes, of which 1,277 span the cutoff. It measures differences in six-month updating, whereas the repeated-cross-section headline measures a change in group levels. The estimate is imprecise around zero, but cannot be interpreted as a low-power replication of the headline or as a bound on composition bias.

Winsorisation. With headline controls and the same 166,125 observations, 1/99 winsorisation gives -1.030 (SE 0.144), while 5/95 winsorisation gives -0.349 (SE 0.093). Both remain negative and significant, but the attenuation is substantial, consistent with the concentration of the unmodified effect in high forecasts.

Table 8: Consolidated robustness

	Coef.	SE	N
<i>Headline</i>			
+Demog. $\times$ Post	-1.784***	0.198	166,125
<i>Sample and composition</i>			
Phone-only (CATI)	-1.892***	0.228	146,801
Cell-reweighted	-2.402***	0.228	164,766
+ Homeownership interaction	-1.618***	0.188	164,766
<i>Alternative timing</i>			
Post = March 2021	-1.709***	0.261	166,125
Post = July 2021	-1.630***	0.256	166,125
Post = January 2022	-2.343***	0.192	166,125
<i>Partisanship</i>			
Triple-diff (full SES)	-2.075***	0.271	77,574
<i>Placebo cutoffs</i>			
2015	-0.098	0.072	58,285
2018	+0.039	0.081	49,319
<i>Winsorization</i>			
1/99	-1.030***	0.144	166,125
5/95	-0.349***	0.093	166,125
<i>Specificity</i>			
Home-price forecast errors (MAE)	-0.181	0.133	93,298
<i>Within-household panel</i>			
Panel DiD	-0.124	0.292	51,664

Notes: Each row reports the $Stock \times Post$ coefficient from a separate regression; the headline +Demog. $\times$ Post row repeats Table 2 for reference. Rows use one-year CPI inflation expectations, except the specificity row (absolute home-price forecast errors), the winsorised-outcome rows and the panel row (six-month changes in expectations). Standard errors are clustered by survey month except for the panel row, which reports HC1 standard errors. The partisan triple-diff is estimated on the POLAFF subsample. ** $p < 0.05$, *** $p < 0.01$.

2.8 Model Calibration and Quantitative Implications

The quantitative exercise asks whether persistent differences in information access can account for changing disagreement as the inflation environment changes. Its contribution is to confront the mechanism with both average forecasts and their dispersion: the same attention choices must explain how far forecasts lie from the financial anchor and how widely they are spread. Attention costs remain constant across the two regimes, while the mean and dispersion of experienced signals can change.

This is a *model-based minimum-distance calibration*. Parameters are chosen to minimise

discrepancies between model-implied and observed moments, subject to the attention rule and the signal structure. The exercise assesses quantitative fit and the restrictions responsible for it. It does not recover uniquely determined household attention or establish a causal effect of stock ownership. The two-regime comparison also does not estimate the accumulation of salience: duration is assessed through the conditional theory and the separate timing evidence in Section 2.5.

2.8.1 Measurement and model restrictions

Let π_t^F denote the monthly Cleveland Fed expected-inflation anchor and define $R_{i,t} = E_{i,t} - \pi_t^F$, the household forecast relative to that anchor. The eight targets are the mean and standard deviation of R for each group before and after 2021. Survey and macro inputs use the same months, with observations in both ownership groups and an observed twelve-month inflation target: 273 months before 2021 and 54 afterwards, ending in July 2025. Survey weights are normalised within month and group; months then receive equal weight. Subtracting the monthly anchor removes its common movement before computing dispersion. The resulting regime dispersion still includes changes in monthly mean residual forecasts; it is not a direct measurement of individual signal noise.

Let π_τ^F , π_τ^E and $\bar{\pi}_\tau$ denote regime averages of the financial anchor, the food-and-energy price proxy and realised inflation over the forecast horizon. The index τ denotes the pre- or post-2021 regime. A fitted adjustment η_τ links the experienced-price proxy to the model's experienced-signal mean. Define

$$a_{E,\tau} = \pi_\tau^E + \eta_\tau - \bar{\pi}_\tau, \quad a_{F,\tau} = \pi_\tau^F - \bar{\pi}_\tau, \quad W_\tau = a_{E,\tau} - a_{F,\tau}.$$

The adjustment η_τ is distinct from the theoretical bias b_τ : the observed proxy itself can differ from the forecasting target. Under the regime approximation, the latent experienced signal relative to the monthly financial anchor has mean W_τ and standard deviation s_τ , shared by the

two groups. Thus the fitted forecast mean is

$$\mu_{g,\tau} = \pi_\tau^F + (1 - m_{g,\tau}^*)W_\tau, \quad E(R_g) = (1 - m_{g,\tau}^*)W_\tau. \quad (2.8.1)$$

The fitted non-stockholder-minus-stockholder gap is

$$\Delta_\tau = (m_{S,\tau}^* - m_{N,\tau}^*)W_\tau, \quad W_\tau = \pi_\tau^E + \eta_\tau - \pi_\tau^F. \quad (2.8.2)$$

Neither expression requires zero non-stockholder attention.

For this benchmark, attention minimises the retrospective quadratic criterion

$$L_g(m) = [ma_{F,\tau} + (1 - m)a_{E,\tau}]^2 + m^2\sigma_F^2 + (1 - m)^2s_\tau^2 + \kappa_g m^2/2,$$

giving

$$m_{g,\tau}^* = \Pi_{[0,1]} \left(\frac{s_\tau^2 + a_{E,\tau}W_\tau}{s_\tau^2 + \sigma_F^2 + W_\tau^2 + \kappa_g/2} \right). \quad (2.8.3)$$

Here $\Pi_{[0,1]}$ truncates to $[0, 1]$. Setting $a_{F,\tau} = 0$, $a_{E,\tau} = b_\tau$ and $s_\tau^2 = \sigma_E^2$ recovers (2.3.5). Substituting realised regime errors is an additional approximation; it does not imply that households anticipated those errors. The theoretical duration argument retains its ex ante unbiased financial signal.

With a common realised financial anchor and attention common within each group, the fitted dispersion of residual forecasts is

$$d_{g,\tau} \equiv \text{SD}(R_g) = (1 - m_{g,\tau}^*)s_\tau. \quad (2.8.4)$$

Financial noise affects the attention criterion but is not treated as independent cross-sectional forecast variation in this benchmark. Reported forecasts are treated as continuous latent beliefs after the stated survey cleaning; reporting and selection are examined in the extensions below. Set $\sigma_F^2 = 0.25$ and maintain $\kappa_N = \kappa_S + \delta$, $\delta > 0$, as the model's information-access restriction. This ordering is an assumption, not a finding of the calibration.

Six parameters— κ_S , δ , W_{pre} , W_{post} , s_{pre} , s_{post} —are fitted to the eight targets by bounded

nonlinear least squares:

$$\hat{\theta} = \arg \min_{\theta} \sum_{\tau \in \{\text{pre}, \text{post}\}} \sum_{g \in \{S, N\}} \left[((1 - m_{g,\tau}^*(\theta))W_{\tau} - \bar{R}_{g,\tau})^2 + (d_{g,\tau}(\theta) - \widehat{SD}(R_{g,\tau}))^2 \right]. \quad (2.8.5)$$

The criterion weights the eight discrepancies equally; survey weights enter the construction of the targets. Having more targets than parameters makes the restrictions assessable, but does not itself prove identification or turn the residual sum of squares into a formal specification test.

2.8.2 What the benchmark explains

Table 9 reports observed and fitted moments. At the reported solution, financial attention is 0.08 before 2021 and 0.38 afterwards for stockholders, and close to zero for non-stockholders. These are conditional model weights. The non-stockholder cost reaches the search ceiling, so a precise finite value is not recovered. Higher experienced-signal mean errors and dispersion increase the incentive to attend at unchanged attention costs.

The fitted gap widens by 2.02 percentage points, compared with 2.09 in these unconditional calibration targets. Yet its level is too small in both regimes: 0.13 versus 0.82 before 2021, and 2.16 versus 2.91 afterwards. The benchmark therefore reproduces much of the change while failing to account fully for the pre-existing group difference. These unconditional comparisons are distinct from the controlled difference-in-differences estimate. Nor is the fitted widening an unused validation target: it is a contrast of the means entering the criterion.

The joint discipline is visible in a restriction that holds before imposing the attention rule. Equations (2.8.1) and (2.8.4) imply

$$\frac{E(R_S)}{E(R_N)} = \frac{d_S}{d_N} = \frac{1 - m_S^*}{1 - m_N^*}. \quad (2.8.6)$$

In the data, means imply relative compression of 0.393 before 2021 and 0.490 afterwards, whereas SDs imply 0.182 and 0.348, respectively. Here compression is one minus the corresponding ratio, and equals $(m_S^* - m_N^*)/(1 - m_N^*)$ under the model. Thus the homogeneous model cannot match all eight point targets, even if both groups' attention weights are freely chosen. This is a

Table 9: Minimum-distance calibration: observed and fitted moments

	Pre-2021		Post-2021	
	Observed	Fitted	Observed	Fitted
Stockholders, mean R	1.27	1.64	3.03	3.49
Non-stockholders, mean R	2.09	1.77	5.94	5.65
Stockholders, SD of R	3.46	3.73	7.24	6.96
Non-stockholders, SD of R	4.23	4.03	11.11	11.26
Gap between group means	0.82	0.13	2.91	2.16
<i>Conditional model quantities</i>				
Stockholder attention m_S	0.08		0.38	
Non-stockholder attention m_N	$\simeq 0$		$\simeq 0$	
Signal difference W	1.77		5.65	
Latent experienced-signal SD s	4.03		11.26	

Notes: R is the reported forecast minus the same month's financial anchor. All means and SDs are in percentage points. Targets use identical survey and macro support, equal month weights and survey weights within month and group: 273 pre-2021 months and 54 post-2021 months through July 2025; 159,367 reports in total. Six parameters target the first four rows in each regime; the gap row is derived. The identity-weighted residual sum of squares is 0.7466. Attention costs are constant across regimes and ordered by assumption. The solution approaches negligible non-stockholder attention; this is not imposed as an exact zero. Numerical ranges and sensitivity exercises are reported in Appendix 2.G.

diagnostic discrepancy, not by itself a statistical rejection. It also explains why dispersion adds substantive discipline beyond fitting group means.

2.8.3 Heterogeneity and sensitivity

Allowing individual attention costs to differ relaxes the common-weight restriction while retaining optimal attention and a cost distribution that is constant across regimes. On the same eight targets, a seven-parameter lognormal-cost extension lowers squared fitting error from 0.7466 to 0.0316; an eight-parameter two-type extension lowers it to 0.0058. Average attention and its variance are both derived from individual choices. The improvement shows that common attention within a group is consequential for quantitative fit. It does not establish cost heterogeneity as the unique source of the discrepancy.

Appendix 2.G reports the comparison and its limits. Alternative latent-signal scales and weighting choices yield substantially different attention paths, and additional distributional checks retain important discrepancies. Explicit reporting and selection, skewed signals, and demographic overlap weighting were also examined. These checks support treating the exercise

as an assessment of the mechanism's quantitative restrictions, rather than selecting a near-exact fit as a uniquely identified structural account. Further determination of absolute attention requires independent information on attention or experienced signals.

The experienced-price proxy provides a measurement reference, but its level does not independently discipline attention when η_τ is freely fitted: replacing π_τ^E by $\pi_\tau^E + c_\tau$ and η_τ by $\eta_\tau - c_\tau$ leaves W_τ and the criterion unchanged. Invariance to that replacement is algebraic, not a separate robustness result. Likewise, (2.8.6) determines relative compression under its assumptions, not either group's absolute attention. Changing the cost function can therefore change fitted attention; quadratic costs remain a maintained behavioural restriction.

2.8.4 Distributional implications

The model links attention to the upper tail through a precise conditional mapping. With common within-group attention, a common financial anchor and the same experienced-signal distribution, define $\lambda_\tau = (m_{S,\tau}^* - m_{N,\tau}^*) / (1 - m_{N,\tau}^*)$. For $m_{N,\tau}^* < 1$, the residual forecast quantiles satisfy

$$Q_q(R_{S,\tau}) = (1 - \lambda_\tau)Q_q(R_{N,\tau}). \quad (2.8.7)$$

Thus the gap is larger at quantiles further above the anchor. Positive non-stockholder attention is allowed. A near-zero lower-tail gap requires those forecasts to lie near the anchor, and a steeper gradient after 2021 requires a change in relative attention or the experienced-signal distribution. Neither follows from a higher experienced-signal mean alone.

This explains a distributional implication of the mechanism; it does not validate the benchmark's numerical fit to the tails. Means and SDs are targeted, whereas additional quantiles can assess restrictions unused in fitting. The paper's controlled quantile-regression estimates remain distinct from this unconditional mapping; the full derivation is in Appendix 2.A.

2.8.5 Interpreting changes in the informed share

A fitted near-zero non-stockholder weight cannot rule out entry into financial attention. The supplementary exercises in Appendix 2.F instead examine whether more non-stockholders report

forecasts close to the financial anchor. Those shares decline during the surge. This is descriptive evidence against the particular prediction of a growing share of reports near the anchor, not a direct observation of information acquisition. Forecast proximity also depends on signal disagreement, noise and reporting. The results therefore do not exclude positive or increasing attention among some non-stockholders, or changes in group composition.

2.9 Conclusion

The gap between stockholders' and non-stockholders' inflation expectations was small and stable for two decades and then more than tripled during the 2021–2025 surge. In the most conservative specification stockholders' expectations rose 1.78 percentage points less than non-stockholders', and the divergence was an improvement in accuracy, not merely a lower number: stockholders' absolute forecast errors and upward bias both fell relative to non-stockholders'. The effect is concentrated in the upper tail: the divergence appears where beliefs are most extreme rather than as a uniform downward shift. The model explains why attention can compress the upper tail under a common financial anchor, attention common within group, and a shared experienced-signal distribution. These restrictions make the distributional evidence informative about the proposed mechanism without establishing it as the sole explanation.

The longer post-2021 inflation episode was accompanied by an expectations gap that developed over time and became larger during disinflation. The timing survives controls for observed composition, month-year fixed effects and a restriction to phone interviews. Together with the much smaller response around 2008, it supports an account in which accumulated price experience and different access to inflation information sustain disagreement. Duration is central to that account, while its causal contribution is not separately identified from shock magnitude, delayed learning and other episode differences. The quantitative assessment connects this mechanism to the observed scale of disagreement. With constant attention costs, the homogeneous benchmark reproduces much of the widening while missing the gap levels. Jointly targeting means and dispersions exposes a restriction that mean matching alone would conceal, and individual cost heterogeneity substantially improves fit. This establishes the quantitative

consequences of the model's assumptions; it does not uniquely recover attention paths or rule out compositional changes.

The effect is larger for women and lower-income households, and gender moderation persists when education and income moderation enter jointly. This pattern is consistent with differences in salient-price exposure, but the observed characteristics do not isolate that channel from attention costs, financial knowledge or reporting behaviour.

The findings suggest a communication question: how can credible aggregate-price information reach households whose expectations remain elevated during disinflation? Financial monitoring offers one possible channel, and the holdings gradient is consistent with attention varying with portfolio stakes. Neither result establishes that expanding stock ownership would reproduce the observed association. The model instead motivates testing information delivery directly, particularly among households with limited financial engagement. This is a mechanism-based implication, not an estimated policy effect.

Two limits define the interpretation. The empirical analysis conditions on observed ownership, and the sensitivity checks do not remove all possible selection. The calibration imposes signal and attention restrictions whose quantitative consequences can be assessed, but the fitted weights are not direct measures of attention. The central empirical finding remains a state-dependent gap that widened during the prolonged post-2021 episode and remained large as inflation receded. The timing makes accumulated experience a substantive explanation to investigate, without identifying duration separately from the episode's other features.

Appendix

2.A Model Derivations

This appendix derives the results stated in Section 2.3: the attention loss (2.3.4), the optimal weight (2.3.5) and its comparative statics, the bundling ranking $m_S^* > m_N^*$, the group means and gap (2.3.6), the duration decomposition, and the tail-compression prediction (2.8.7). Throughout, a household in group $g \in \{S, N\}$ has attention cost κ_g , and the signal noises $\varepsilon^F, \varepsilon^E$ are independent, mean zero, with variances σ_F^2, σ_E^2 .

2.A.1 Forecast error and effective signal noise

Substituting the signals (2.3.1) into the forecast rule (2.3.2), the forecast error decomposes into a deterministic bias term and two noise terms,

$$E_i - \pi = (1 - m) b_\tau + m \varepsilon_i^F + (1 - m) \varepsilon_i^E. \quad (2.A.1)$$

The forecast is biased by $(1 - m)b_\tau$: attending more to the financial signal (higher m) carries less of the experienced-price bias. Squaring (2.A.1) and taking expectations, the cross terms vanish because the noises are independent and mean zero and b_τ is deterministic, leaving the mean squared error

$$\mathbb{E}[(E_i - \pi)^2] = (1 - m)^2 b_\tau^2 + m^2 \sigma_F^2 + (1 - m)^2 \sigma_E^2 = (1 - m)^2 \Sigma_{E,\tau}^2 + m^2 \sigma_F^2, \quad (2.A.2)$$

where $\Sigma_{E,\tau}^2 \equiv \sigma_E^2 + b_\tau^2$. Bias and variance enter the loss symmetrically, as the two components of mean squared error; neither alone determines how useful the signal is for forecasting. Adding the quadratic attention cost $\frac{\kappa_g}{2} m^2$ yields the objective (2.3.4).

2.A.2 Optimal attention weight

Write the objective as $O(m) = (1 - m)^2 \Sigma_{E,\tau}^2 + m^2 \sigma_F^2 + \frac{\kappa_g}{2} m^2$. Its first derivative is

$$O'(m) = -2(1 - m) \Sigma_{E,\tau}^2 + 2m \sigma_F^2 + \kappa_g m = 0, \quad (2.A.3)$$

which rearranges to $\Sigma_{E,\tau}^2 = m(\sigma_F^2 + \Sigma_{E,\tau}^2 + \kappa_g/2)$ and hence to the optimal weight (2.3.5),

$$m_{g,\tau}^* = \frac{\Sigma_{E,\tau}^2}{\sigma_F^2 + \Sigma_{E,\tau}^2 + \kappa_g/2}. \quad (2.A.4)$$

The second derivative $O''(m) = 2\Sigma_{E,\tau}^2 + 2\sigma_F^2 + \kappa_g > 0$, so O is strictly convex and (2.A.4) is its unique minimiser. Since the numerator is positive and strictly below the denominator, $0 < m_{g,\tau}^* < 1$ for every finite κ_g . Two limits bound it: as $\kappa_g \rightarrow \infty$, $m_{g,\tau}^* \rightarrow 0$ —the non-stockholder corner, where attention is too costly to use the financial signal at all—and as $\kappa_g \rightarrow 0$, $m_{g,\tau}^* \rightarrow \Sigma_{E,\tau}^2/(\sigma_F^2 + \Sigma_{E,\tau}^2)$. This is the weight a household places when attention is free, and it is *not* the Bayesian precision weight, which is $\sigma_E^2/(\sigma_F^2 + \sigma_E^2)$ and involves no bias term: the two agree only at $b_\tau = 0$. The limit here carries b^2 because the household weighs the experienced signal by its full mean squared error rather than by its precision, which is the behavioural restriction of Section 2.3.2, not a posterior.

2.A.3 Comparative statics and the bundling ranking

Let $D_{g,\tau} \equiv \sigma_F^2 + \Sigma_{E,\tau}^2 + \kappa_g/2$ be the denominator of (2.A.4). Differentiating,

$$\frac{\partial m_{g,\tau}^*}{\partial \kappa_g} = -\frac{\Sigma_{E,\tau}^2}{2 D_{g,\tau}^2} < 0, \quad \frac{\partial m_{g,\tau}^*}{\partial \Sigma_{E,\tau}^2} = \frac{\sigma_F^2 + \kappa_g/2}{D_{g,\tau}^2} > 0. \quad (2.A.5)$$

Attention falls with its cost and rises as the experienced signal deteriorates. The bundling discount (2.3.3), $\kappa_S = \kappa_N - \delta$ with $\delta > 0$, then gives $m_{S,\tau}^* > m_{N,\tau}^*$ at every date, because m^* is decreasing in κ .

2.A.4 Group means and the cross-sectional gap

With mean-zero signal noises and attention common within a group, taking expectations in (2.3.2) gives $\mu_{g,\tau} = \pi + (1 - m_{g,\tau}^*)b_\tau$. Hence

$$\Delta_\tau \equiv \mu_{N,\tau} - \mu_{S,\tau} = (m_{S,\tau}^* - m_{N,\tau}^*)b_\tau, \quad (2.A.6)$$

which is the general baseline gap in (2.3.6). Setting $m_{N,\tau}^* = 0$ gives the special case $\Delta_\tau = m_{S,\tau}^*b_\tau$. In this baseline, $\mu_{N,\tau} - \pi = (1 - m_{N,\tau}^*)b_\tau$: the non-stockholder mean error equals b_τ only at zero financial attention and otherwise understates a positive b_τ . The empirical implementation in Section 2.8 distinguishes errors relative to the forecast target from residuals relative to observed price proxies.

To establish the large-bias limit, hold both finite attention costs and signal noise variances fixed. Write $\Sigma_{E,\tau}^2 = \sigma_E^2 + b_\tau^2$ and $D_{g,\tau} = \sigma_F^2 + \Sigma_{E,\tau}^2 + \kappa_g/2$. Using (2.3.5) and $\delta = \kappa_N - \kappa_S > 0$ gives

$$\Delta_\tau = \frac{\delta b_\tau \Sigma_{E,\tau}^2}{2D_{S,\tau}D_{N,\tau}}, \quad \lim_{b_\tau \rightarrow \infty} b_\tau \Delta_\tau = \frac{\delta}{2}.$$

Thus $\Delta_\tau \sim \delta/(2b_\tau) \rightarrow 0$, even though each attention weight approaches one. This limit describes the baseline model with fixed finite costs; it neither locates an empirical turning point nor establishes that the sample reaches the declining part of the gap.

2.A.5 Duration: experienced-signal bias and relative attention

Maintain the baseline signals in (2.3.1), including zero financial mean bias, and let $b_\tau = b_0 + \gamma S_\tau$. Both groups share these mean signals and $\gamma > 0$. Hold attention costs and noise variances fixed. Define

$$D_{g,\tau} = \sigma_E^2 + b_\tau^2 + \sigma_F^2 + \kappa_g/2.$$

The baseline attention rule (2.3.5) gives

$$m_{g,\tau}^* = \frac{\sigma_E^2 + b_\tau^2}{D_{g,\tau}}, \quad \frac{\partial m_{g,\tau}^*}{\partial b_\tau} = \frac{2b_\tau(\sigma_F^2 + \kappa_g/2)}{D_{g,\tau}^2}.$$

Writing $A_\tau = m_{S,\tau}^* - m_{N,\tau}^*$ and applying the product and chain rules to $\Delta_\tau = A_\tau b_\tau$ yields

$$\frac{\partial \Delta_\tau}{\partial S_\tau} = \gamma \left[A_\tau + b_\tau \left(\frac{\partial m_{S,\tau}^*}{\partial b_\tau} - \frac{\partial m_{N,\tau}^*}{\partial b_\tau} \right) \right]. \quad (2.A.7)$$

For $b_\tau > 0$ and $A_\tau > 0$, the gap widens if and only if

$$\frac{\partial A_\tau}{\partial b_\tau} > -\frac{A_\tau}{b_\tau}. \quad (2.A.8)$$

A non-decreasing attention difference is sufficient, but not necessary: the gap can also widen while that difference shrinks, provided the decline is sufficiently small. Each group's attention increases with b_τ , but the difference between their attention weights need not increase. These results allow positive attention in both groups.

2.A.6 Tail compression

Condition on a common realised financial signal s_τ^F and a shared experienced-signal distribution $s_{i,\tau}^E$ across the two groups. Attention is common within each group, with $0 \leq m_{N,\tau}^* < m_{S,\tau}^* \leq 1$. These are additional distributional restrictions on (2.3.2). The forecast in group g is

$$E_{i,\tau}^g = m_{g,\tau}^* s_\tau^F + (1 - m_{g,\tau}^*) s_{i,\tau}^E.$$

Define $A_\tau = m_{S,\tau}^* - m_{N,\tau}^*$ and $\lambda_\tau = A_\tau / (1 - m_{N,\tau}^*)$. Eliminating the shared experienced signal gives an affine relation in distribution with non-negative slope. Therefore

$$Q_q(E_\tau^S) = \lambda_\tau s_\tau^F + (1 - \lambda_\tau) Q_q(E_\tau^N). \quad (2.A.9)$$

Subtracting the common financial anchor gives the residual-forecast mapping (2.8.7). When $m_{N,\tau}^* = 0$, $\lambda_\tau = m_{S,\tau}^*$. In general the quantile gap is

$$\begin{aligned}\Delta_{q,\tau} &\equiv Q_q(E_\tau^N) - Q_q(E_\tau^S) \\ &= A_\tau [Q_q(s_\tau^E) - s_\tau^F] \\ &= \lambda_\tau [Q_q(E_\tau^N) - s_\tau^F].\end{aligned}\tag{2.A.10}$$

It is increasing across quantiles and positive where the experienced-signal quantile exceeds the financial anchor. Financial attention can raise low forecasts that lie below the anchor; a near-zero lower-tail gap requires those forecasts to lie near it.

The within-period gradient alone does not establish a gradient in changes between periods. Under the Gaussian experienced-signal assumption, write $s_{i,\tau}^E = \mu_\tau^E + \sigma_{E,\tau} Z_i$, where Z_i is standard normal, and let z_q denote its q th quantile. For $q_H > q_L$,

$$\Delta_{q_H,\tau} - \Delta_{q_L,\tau} = A_\tau \sigma_{E,\tau} (z_{q_H} - z_{q_L}).$$

A steeper post-period gradient therefore requires $A_{\text{post}} \sigma_{E,\text{post}} > A_{\text{pre}} \sigma_{E,\text{pre}}$. Because the empirical stockholder-minus-non-stockholder contrast has the opposite sign, this condition predicts a more negative change at higher quantiles. A uniform rise in b_τ shifts the experienced-signal distribution but does not spread it; holding attention and noise dispersion fixed, that shift leaves the quantile gradient unchanged. Conditional quantile-regression coefficients and this distributional mapping are distinct estimands, so the comparison in Section 2.6 assesses consistency with the mechanism.

2.A.7 Heterogeneity in γ

To isolate salience sensitivity, compare otherwise identical households at the two group attention costs, allowing their common sensitivity γ to vary across comparisons. Hold S_τ , attention costs, noise variances and the baseline unbiased financial signal fixed. Write

$$b_\tau(\gamma) = b_0 + \gamma S_\tau, \quad \Delta_\tau(\gamma) = A_\tau(b_\tau(\gamma)) b_\tau(\gamma).$$

Here $A_\tau(b)$ is the difference in optimal attention at the two costs. Applying the chain rule gives

$$\frac{\partial \Delta_\tau(\gamma)}{\partial \gamma} = S_\tau \left[A_\tau + b_\tau \frac{\partial A_\tau}{\partial b_\tau} \right]. \quad (2.A.11)$$

For $S_\tau > 0$, $b_\tau > 0$ and $A_\tau > 0$, the gap increases with sensitivity under exactly the positive-response condition (2.A.8). Holding attention weights fixed gives the direct effect $S_\tau A_\tau > 0$; when attention responds, convergence between the weights can offset it.

This is a comparison within the model, not an identification of household sensitivities from observed group differences. Shopping frequency and the salience of purchases may influence γ_i , but the survey does not measure these channels directly. Income, education, age and gender provide imperfect indicators and may also reflect attention costs or analytical capacity. Their interactions assess observable heterogeneity; they do not separately recover γ_i or establish that the gradient operates through exposure rather than other household differences.

2.B Full Specification Coefficients

Table 10: Full coefficients, headline specification (+Demog. $\times$ Post)

	Coefficient
Stockholder	−0.479*** (0.028)
Stock $\times$ Post	−1.784*** (0.198)
Female	+0.593*** (0.024)
Female $\times$ Post	+0.912*** (0.189)
Under 45	−0.171*** (0.026)
Under 45 $\times$ Post	+0.164 (0.133)
College graduate	−0.404*** (0.027)
College $\times$ Post	−1.254*** (0.129)
High income	−0.483*** (0.025)
High income $\times$ Post	−0.570*** (0.147)
Education missing	−0.422** (0.172)
Education missing $\times$ Post	+0.329 (0.583)
Income missing	−0.213*** (0.052)
Income missing $\times$ Post	−0.176 (0.286)
Age missing	+0.029 (0.177)
Age missing $\times$ Post	−0.510 (0.595)
Sex missing	−0.537 (0.865)

Notes: Coefficients from the headline +Demog. $\times$ Post specification (the fourth column of Table 2). Dependent variable: cleaned one-year-ahead inflation expectation. Survey-weighted estimates; standard errors clustered by survey month in parentheses. Survey-year fixed effects (1997–2025) are included but suppressed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2.C Heterogeneity: Full Coefficients

Table 11: Heterogeneity triple-difference estimates, full coefficients

	Education	Income	Gender	Age
Stockholder	−0.492*** (0.035)	−0.523*** (0.034)	−0.450*** (0.036)	−0.450*** (0.037)
Stock × Post	−2.079*** (0.276)	−1.951*** (0.232)	−1.160*** (0.192)	−2.005*** (0.234)
College graduate	−0.424*** (0.047)			
College graduate × Post	−1.758*** (0.292)			
Stock × College graduate	+0.031 (0.052)			
Stock × Post × College graduate	+0.710** (0.361)			
High income		−0.581*** (0.050)		
High income × Post		−1.440*** (0.446)		
Stock × High income		+0.135** (0.058)		
Stock × Post × High income		+1.015** (0.410)		
Female			+0.629*** (0.044)	
Female × Post			+1.857*** (0.321)	
Stock × Female			−0.057 (0.052)	
Stock × Post × Female			−1.329*** (0.322)	
Under 45				−0.124** (0.049)
Under 45 × Post				−0.220 (0.290)
Stock × Under 45				−0.075 (0.057)
Stock × Post × Under 45				+0.552* (0.306)

Notes: Each column is a separate triple-difference regression adding the full set of two-way and three-way interactions of *Stock*, *Post*, and the demographic characteristic to the headline specification; the coefficient of interest is *Stock* × *Post* × characteristic. Dependent variable: one-year-ahead inflation expectation. Survey-weighted; standard errors clustered by survey month in parentheses. ** $p < 0.05$, *** $p < 0.01$.

2.D Forecast Accuracy: Full Coefficients

Table 12: Forecast accuracy, full coefficients

	Absolute error	Signed error
Stock $\times$ Post	−1.381*** (0.181)	−1.497*** (0.188)
Stockholder	−0.470*** (0.024)	−0.494*** (0.028)
Female	0.447*** (0.020)	0.592*** (0.024)
Female $\times$ Post	0.947*** (0.173)	0.727*** (0.199)
Young (< 45)	−0.068*** (0.021)	−0.175*** (0.026)
Young $\times$ Post	−0.015 (0.095)	0.159 (0.144)
College graduate	−0.364*** (0.022)	−0.399*** (0.028)
College $\times$ Post	−0.920*** (0.135)	−1.314*** (0.133)
High income	−0.298*** (0.023)	−0.475*** (0.025)
High income $\times$ Post	−0.605*** (0.164)	−0.534*** (0.171)
Age missing	0.366*** (0.140)	0.042 (0.180)
Age missing $\times$ Post	−0.900* (0.521)	−0.576 (0.632)
Education missing	−0.223* (0.128)	−0.379** (0.179)
Education missing $\times$ Post	0.326 (0.554)	0.472 (0.658)
Income missing	−0.004 (0.042)	−0.215*** (0.053)
Income missing $\times$ Post	−0.306 (0.274)	0.041 (0.307)
Gender missing (post-period)	0.140 (0.850)	−0.683 (1.034)
Constant	3.846*** (0.168)	0.309*** (0.089)
Observations	159,954	159,954
Survey-year FE	Yes	Yes

Notes: Coefficients for the regressions in Table 5, excluding year-effect coefficients. Outcomes are expectation minus realised twelve-month inflation and its absolute value. Survey-weighted estimates with standard errors clustered by survey month. Missing-category indicators and their post interactions are included. Gender nonresponse occurs only after 2021 in this sample, so its indicator and post interaction coincide; a single column represents both. Frozen CPI inputs, complete monthly calendar and no filling of missing observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2.E Event-Study Coefficients

Table 13: Event-study coefficients ($\text{Stock} \times \text{year}$), 2019 base

Year	Coefficient	SE
2007	−0.359**	0.159
2008	−0.152	0.167
2009	−0.171	0.197
2010	−0.487***	0.120
2011	−0.068	0.167
2012	−0.107	0.143
2013	−0.164	0.137
2014	−0.303*	0.183
2015	−0.154	0.139
2016	−0.302***	0.116
2017	−0.126	0.123
2018	−0.054	0.135
2019	— (base)	—
2020	+0.016	0.152
2021	+0.343	0.238
2022	−1.322***	0.383
2023	−2.316***	0.286
2024	−2.382***	0.302

Notes: Coefficients on $\text{Stock} \times \text{year}$ interactions from the difference-in-differences specification, normalised to the 2019 base year; these are the estimates plotted in Figure 2. Survey-weighted; standard errors clustered by survey month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2.F Forecast Proximity and Potential Attention Entry

These exercises examine forecast proximity to the financial anchor and exploratory mixture representations. Neither proximity nor a fitted component share directly measures attention. A near-zero calibration weight does not reject entry, and the following diagnostics do not rule it out generally.

A structural mixture model. Rather than fixing non-stockholders at the corner, this specification lets an unobserved *informed share* among non-stockholders, households that weight the financial signal, vary freely across four sub-periods (pre-2021, 2021, 2022, 2023–25), and estimates that share by maximum likelihood from the group’s expectation distribution. Entry predicts a share that climbs monotonically with duration. With food-and-energy as the experienced signal, the estimated share is instead non-monotone: 58, 79, 68, and 46 percent across the four windows, rising then falling rather than accumulating. The accompanying bias estimates swing from near zero to 17 pp, and their bootstrap confidence intervals exclude their own point estimates. This instability prevents a reliable interpretation of the components as informed and uninformed households; it does not itself diagnose label switching or reject entry. A headline-CPI variant produces a V-shaped share (88, 30, 34, 67 percent) with similarly unstable bias. The alternative proxy therefore does not resolve the interpretation of the fitted components.

Alternative informed-band definitions. The direct heterogeneity test classifies a household as ‘informed’ when its expectation falls within a fixed band of the professional signal. To show the result is not an artifact of that band, the test is repeated for four band widths and three demographic splits (income, gender, education). In every band the informed share falls post-2021, and the only difference-in-differences interactions that reach significance run the *wrong* way for entry. Women’s informed share falls *more*, not less. No band or split produces a steadily increasing share of forecasts near the anchor. Attention could nevertheless increase without producing that pattern if the experienced signal or reporting behaviour changes.

The proximity diagnostics document declining shares near the anchor, while the mixture exercise does not recover a stable informed component. These findings leave actual transitions into attention unmeasured. They cannot exclude positive or increasing attention among some non-stockholders or changes in population composition.

2.G Calibration Restrictions and Extensions

2.G.1 Numerical implementation and scope

The benchmark in Section 2.8 uses positive parameters represented in logs, with $\kappa_N = \kappa_S + \delta$. The search ranges are $\kappa_S \in [e^{-10}, e^{22}]$, $\delta \in [e^{-10}, e^{28}]$, $W_\tau \in [0.03, 150]$ and $s_\tau \in [0.1, 80]$. The cost difference δ reaches its upper bound in the reported solution, so the fit does not determine a finite value for that parameter. Multiple starts and numerical refinements locate candidate minima; they do not prove global uniqueness. All targets and signal inputs share the same monthly support and weighting convention.

The identity-weighted criterion treats equal percentage-point discrepancies in means and SDs equally. Joint calendar-block resampling, with 6-, 12- and 18-month blocks, was used to assess target uncertainty; precision-weighted fits were also examined. Because cost parameters approach boundaries and expanded searches find alternative solutions, no conventional chi-square interpretation is attached to the minimum criterion, and no unrestricted confidence intervals for attention are reported from refits confined to a single solution. The short post-period also limits inference from resampling.

2.G.2 Individual cost heterogeneity

Let household costs κ_i be independent of experienced-signal innovations, and obtain each m_i from (2.8.3). Write $\bar{m}_g = E_g(m_i)$ and $v_g = \text{Var}_g(m_i)$. With the same latent experienced-signal mean W and variance s^2 across groups and a common financial anchor, residual forecasts satisfy $R_i = (1 - m_i)(W + \varepsilon_i)$, where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = s^2$. Independence gives

$$E(R_g) = (1 - \bar{m}_g)W, \quad \text{Var}(R_g) = (1 - \bar{m}_g)^2 s^2 + v_g(s^2 + W^2). \quad (2.G.1)$$

The additional variance term relaxes the homogeneous compression restriction. It is implied by the distribution of optimal attention, rather than fitted as an independent variance parameter.

The lognormal specification sets $\kappa_S(z) = \exp(\ell + \omega z)$, $z \sim \mathcal{N}(0, 1)$, and $\kappa_N(z) = \kappa_S(z) + \delta$. It adds one parameter to the benchmark. The two-type specification assigns probabilities q and

$1 - q$ to κ_L and $\kappa_H > \kappa_L$ for stockholders, with the same additive δ for non-stockholders. It has eight fitted parameters in total. Both specifications keep the cost distribution fixed across regimes and impose the cost ordering. Gaussian quadrature and direct finite mixtures, respectively, integrate individual attention choices.

Table 14: Cost heterogeneity: comparison on identical targets

Specification	Parameters	Squared fitting error
Common cost within each group	6	0.7466
Lognormal costs, additive difference	7	0.0316
Two cost types, additive difference	8	0.0058

Notes: All rows use the same eight means and SDs of anchor-relative forecasts, support and weights as Table 9, and the same retrospective attention criterion with $\sigma_F^2 = 0.25$. These are identity-weighted fits. The two-type specification has as many parameters as targets. Lower fitting error is not an independent test, evidence of unique identification, or validation of unused distributional moments.

2.G.3 Additional checks

Parameter profiles and expanded bounds reveal a trade-off between latent signal magnitudes and attention. Some solutions combine very large experienced signals with high attention in both groups; precision-weighted searches can then produce a falling stockholder attention path despite a widening group gap. These solutions expose weak determination of hidden quantities, rather than providing credible alternative behavioural estimates. Positive non-stockholder attention is also compatible with small changes in fit.

An alternative ex ante formulation assumes that the inflation target equals the financial anchor plus an unpredictable common surprise, uncorrelated with individual signal errors. Its variance enters expected loss independently of attention, leaving the baseline attention rule. This removes the use of future realised errors from the choice problem under an explicit additional structure; it does not resolve the weak determination of latent signal scales.

Exploratory distributional checks examined Gaussian and shifted-Gamma experienced signals, integer reporting and selection into the retained response range $[-5, 50]$. The two-type specification with shifted-Gamma experienced signals used ten parameters to target twelve means, SDs and skewness statistics, reserving quantiles and fourth moments for further assessment. It

retained material discrepancies. For example, its stockholder 90th percentile was 10 versus 7 observed before 2021 and 16 versus 13 afterwards. These are unconditional forecast quantiles, not quantile-regression effects. The signal-shape extensions were introduced after examining initial failures; they are exploratory checks, not prespecified confirmation.

Finally, within-month overlap weighting equalised the joint distribution of education, income, age and gender, including separate missing categories. The discrepancy between relative compression implied by means and SDs became smaller but remained. This changes the comparison population and balances observed characteristics; it does not remove unobserved differences or identify an ownership effect. Taken together, the checks motivate heterogeneous attention as an extension while leaving its absolute scale and a fully satisfactory account of the forecast distribution unresolved.

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